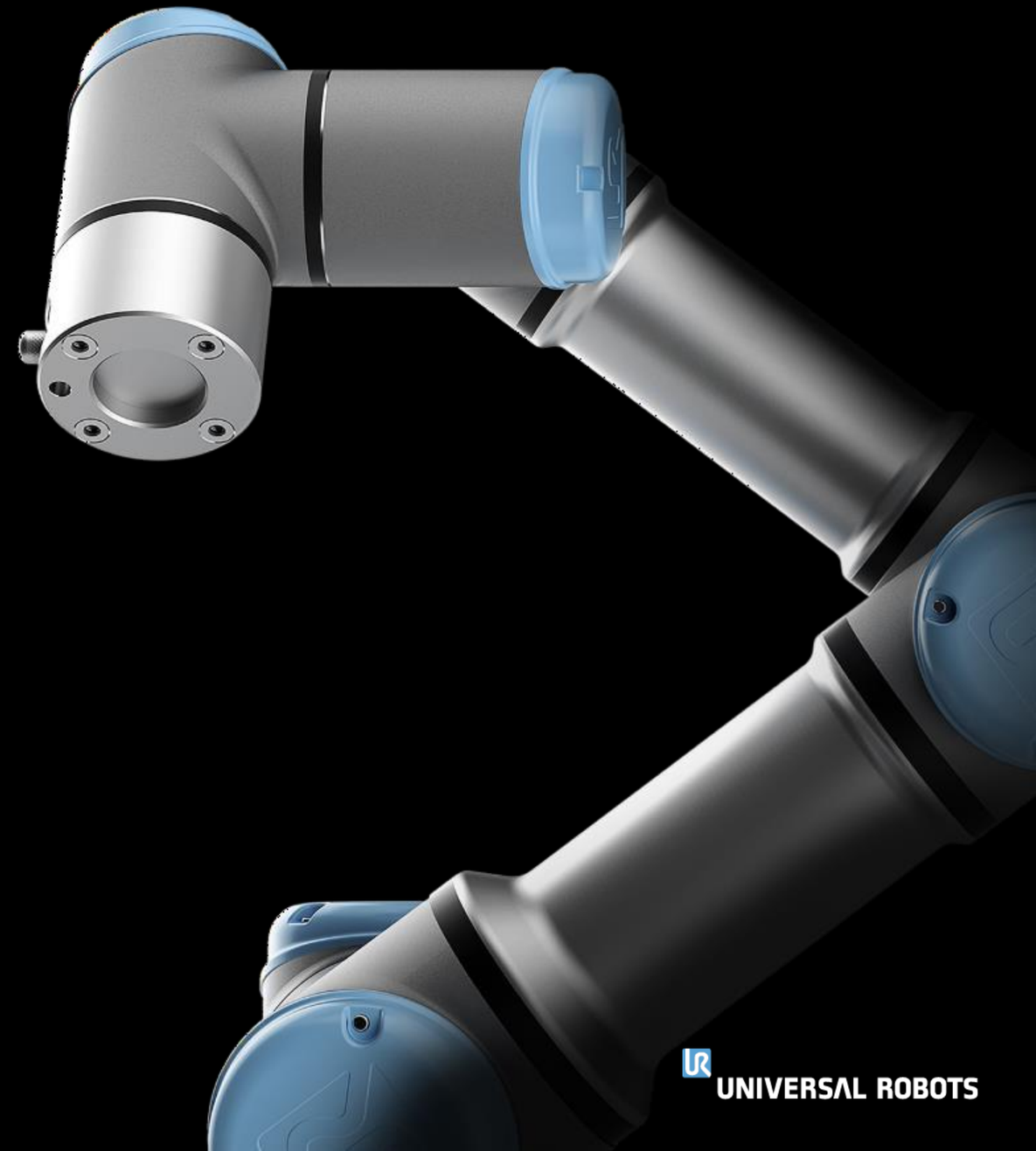
UR DEPLOYMENT GUIDELINES

May 2024



UNIVERSAL ROBOTS

OBJECTIVE

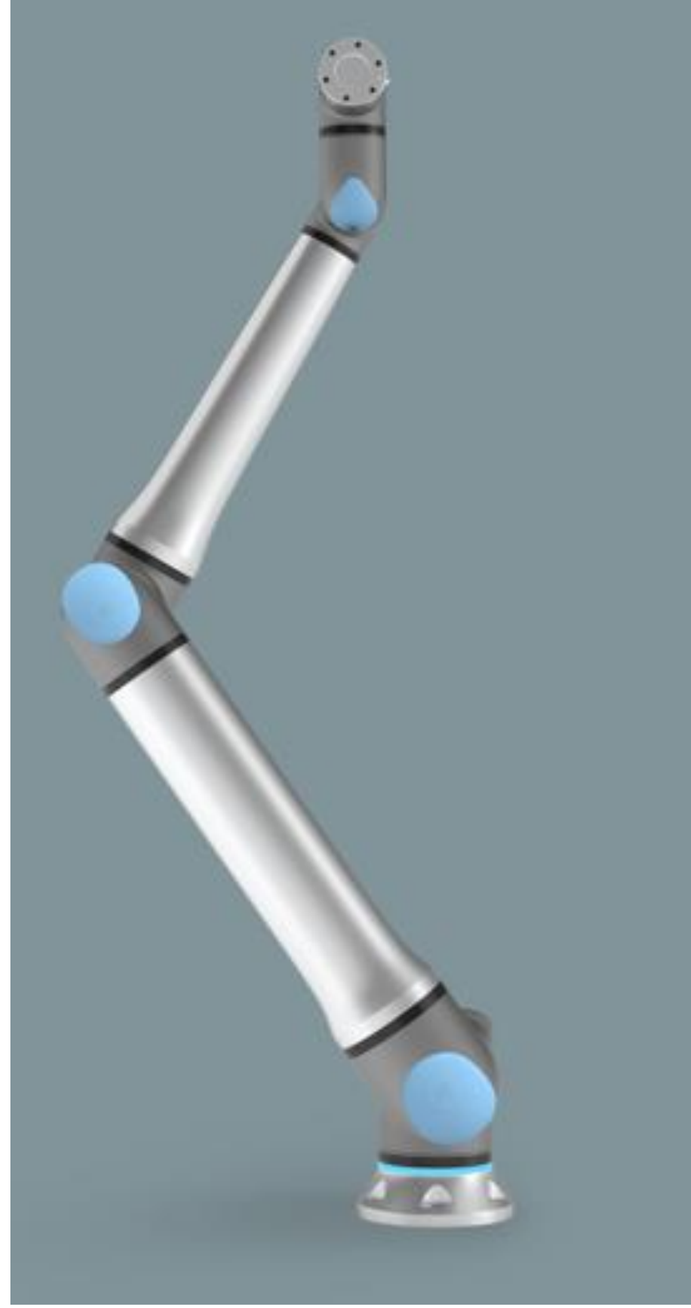
- Understand the most important points to follow to ensure that a UR robot is deployed successfully, in a manner that will ensure a long productive robot lifetime.
- Unless specifically identified otherwise, all items apply to both E-series and newer robots as well as newest release of Polyscope
- The items in this workshop must be communicated to all users programming Universal Robots.



UNIVERSAL ROBOTS

CONTENTS

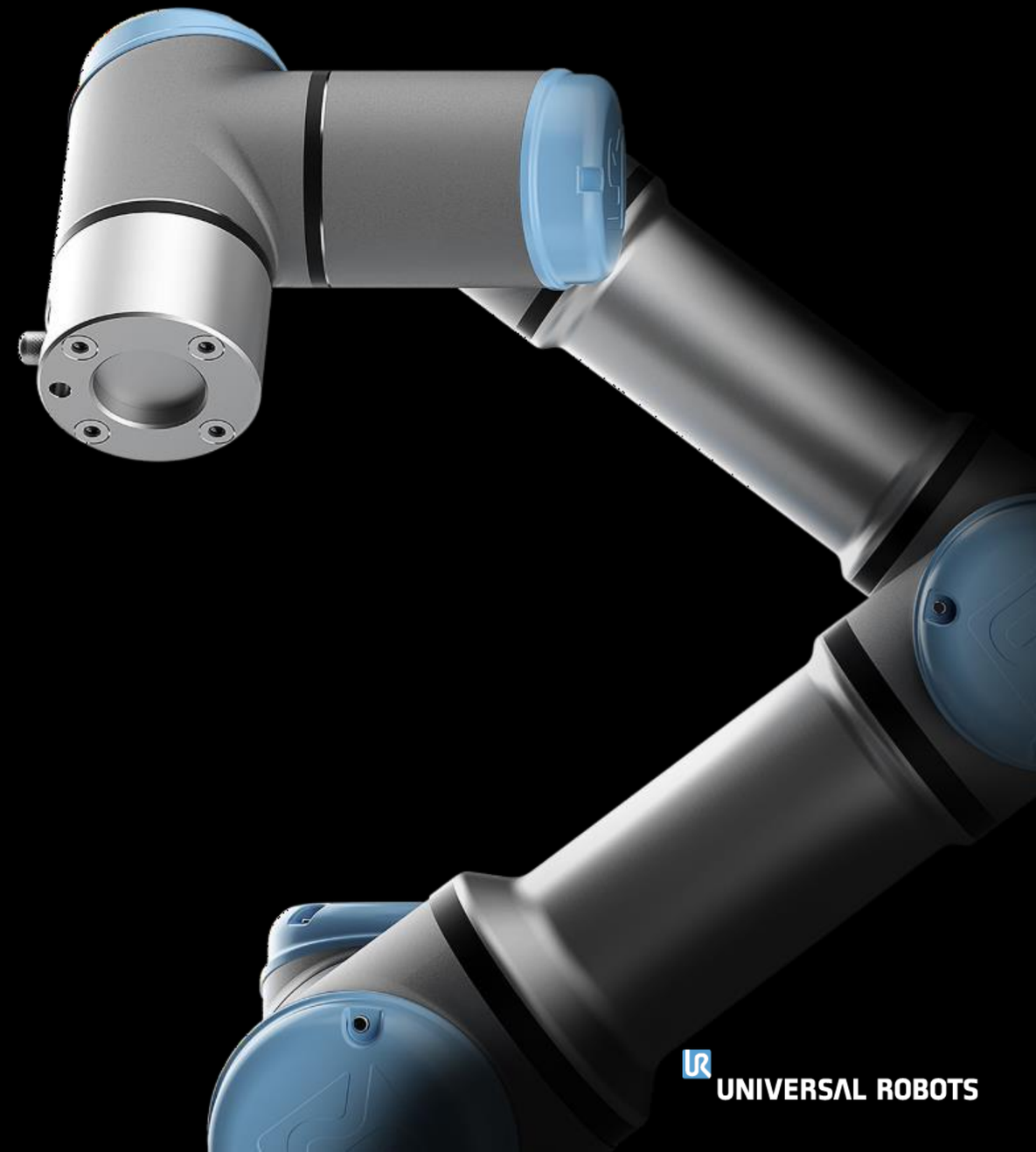
- Installation
  - Mounting Structure
  - Protection from Environment
  - Payload Configuration
- Motion Programming
  - Blend Radii
  - Acceleration Values
  - Singularities
  - Protective Stop
- Maintenance and Troubleshooting
  - Recommended maintenance
  - Troubleshooting strategies



UNIVERSAL ROBOTS

INSTALLATION

Mounting Structure

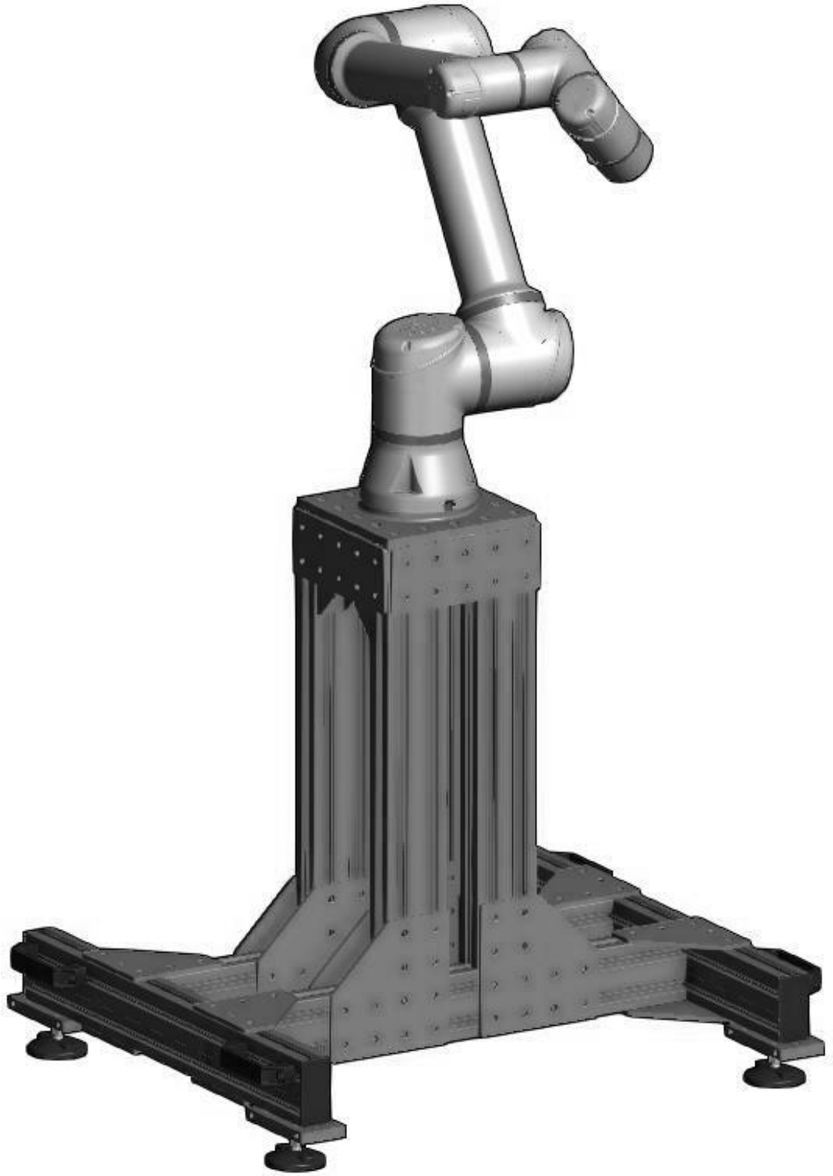


STATIC MOUNTING

- Mounting surface must be sufficiently rigid for robot to operate in a precise repeatable manner

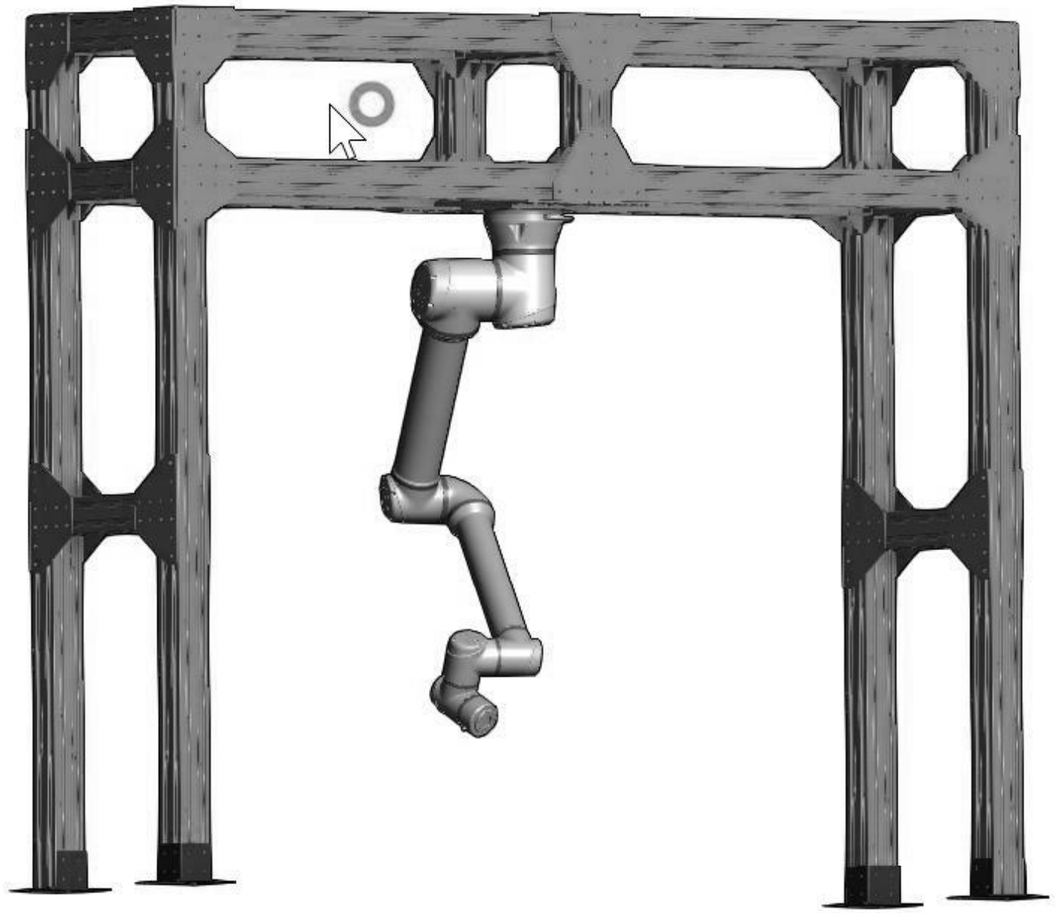
- Must be able to support at least 5 times weight of robot arm

- Must be able to withstand at least ten times the full torque of the robot base joint torques listed below:
  - UR3e: 56Nm
  - UR5e: 150Nm
  - UR10e: 330Nm
  - UR16e: 330Nm
  - UR20/30: 738Nm



INVERTED MOUNTING

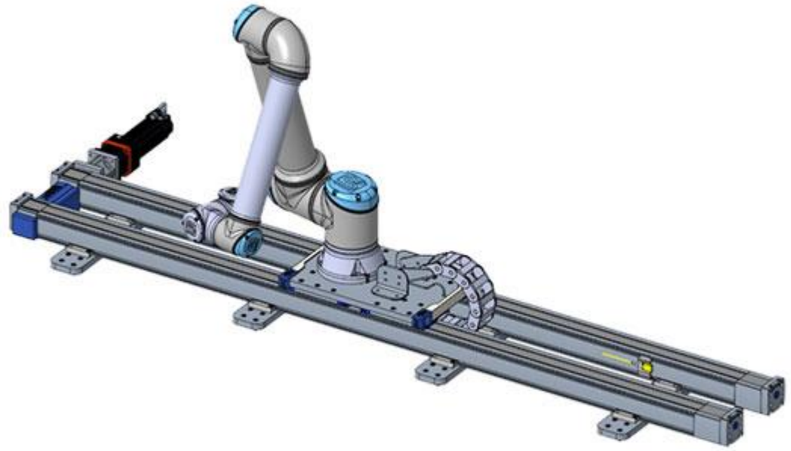
- Same requirements apply to inverted as standard upright mounting.
- Mounting frame must be rigid enough to allow the robot to operate at full speed/payload.
- Remember to configure mounting orientation in installation tab.



UNIVERSAL ROBOTS

7TH AXIS MOUNTING

- Same requirements apply to inverted as standard upright mounting.
- Linear actuator must be controlled in closed loop manner, with full control over speed and acceleration.
- Accelerations must be kept very low, such that the robot cannot is not impacted by the movements.



7TH AXIS MOUNTING-ADDITIONAL CONSIDERATIONS

- If mounted to a liftkit:
  - Refer to the spec sheet to get specific instructions with regards to mounting

- If mounted on a 7th axis slide:
  - To avoid protective stop or wrong mounting faults, ensure that external forces are represented as additional forces using command: set_gravity(gx,gy,gz)



With no external forces applied to a robot mounted in a base down orientation, the command that is issued internally as part of the installation file is:

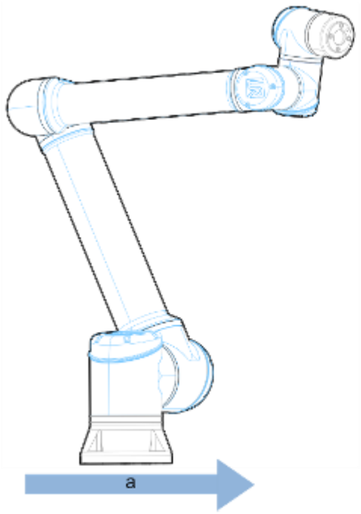
`set_gravity(0.0,0.0,9.82)` because 1g of gravity = 9.82 m/s^2.

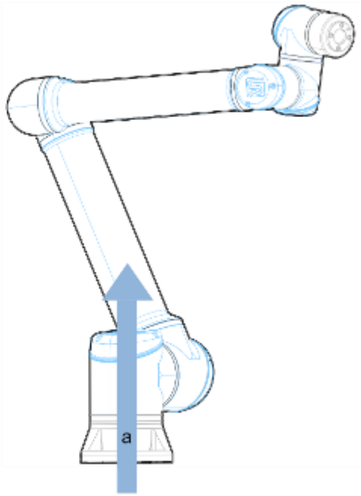
When accelerating the same robot in the x direction with an acceleration rate of ax = 0.5 g or 4.91 m/s^2 the command looks like this:

`set_gravity(4.91,0.0,9.82)`

7TH AXIS ADDITIONAL CONSIDERATIONS

- If the control system senses an unexpected acceleration of the robot joints, a protective stop might result.
    - applicable for both linear and vertical slides/lifts
- The *set_base_acceleration* script function can be used to inform the control system that the robot base is moving, and so prevent unwanted protective stops
- Sets the acceleration of the robot base. This function is used when the robot is attached to a moving base such a linear rail or vertical lift.





Parameters
a: a 3D vector, describing the base acceleration, relative to the base coordinate system of the robot

TOOL MOUNTING

20 KG MAX ⚠ 20 KG MAX ⚠ 20 KG MAX ⚠ 20 KG MAX ⚠ 20 KG MAX ⚠ 20 KG MAX ⚠

- E-series
  - The tool output flange is (ISO 9409-1)
  - Very long M6 bolts can press against the bottom of the tool flange and short circuit the robot.
  - Refer to user manual for spec of maximum bolt length to mount tool
- UR20
  - The tool output flange is (ISO 9409-1-80-6-M8)
  - The tool flange adapter is a mechanical accessory for the tool flange that reduces the tool flange screw pattern from EN ISO-9409-1-80-6-M8 to EN ISO 9409-1-50-4-M6
    - The adapter is limited to a payload of max. 20kg
  - Do not use bolts that extend beyond 10mm to mount the tool
- UR30
  - The tool output flange is (ISO 9409-1-80-6-M8)
  - CANNOT use tool flange adapter that UR20 uses



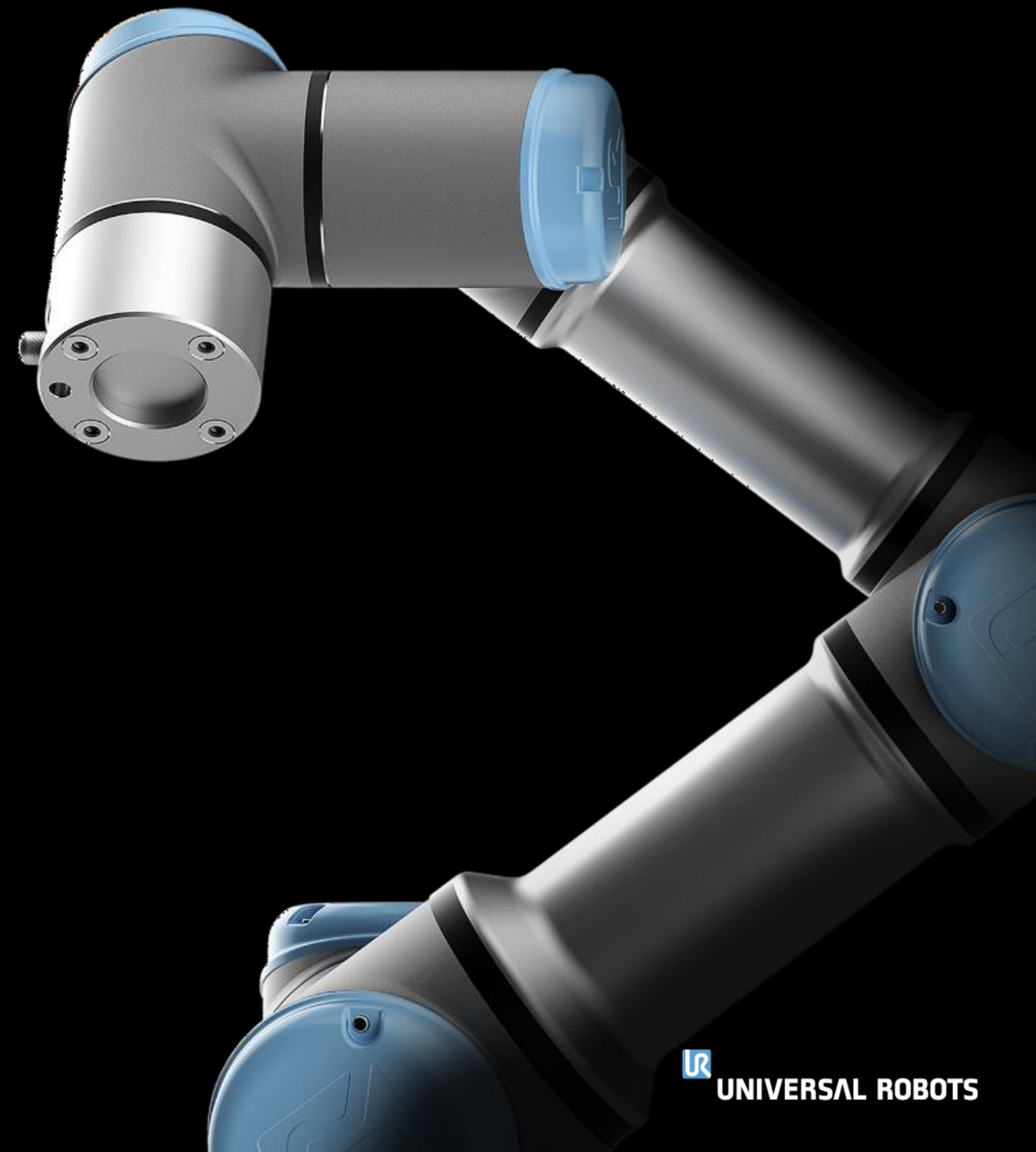
Tool Flange Adapter (UR20) Properties

Interface (robot side)	EN ISO 9409-1-80-6-M8
Interface (end effector side)	EN ISO 9409-1-50-4-M6
Maximum payload	20 kg
Screws (robot side)	6 x M8 8.8 screws tightened to 16 Nm
Screws (end effector side)	4 x M6 8.8 screws tightened to 8 Nm
Positioning (end effector side)	ISO 2338 Ø6 h8
Material	Anodized aluminum
Weight	332 g
Center of mass (x, y, z)	0.00 mm, -0.47 mm, 4.32 mm

UNIVERSAL ROBOTS

INSTALLATION

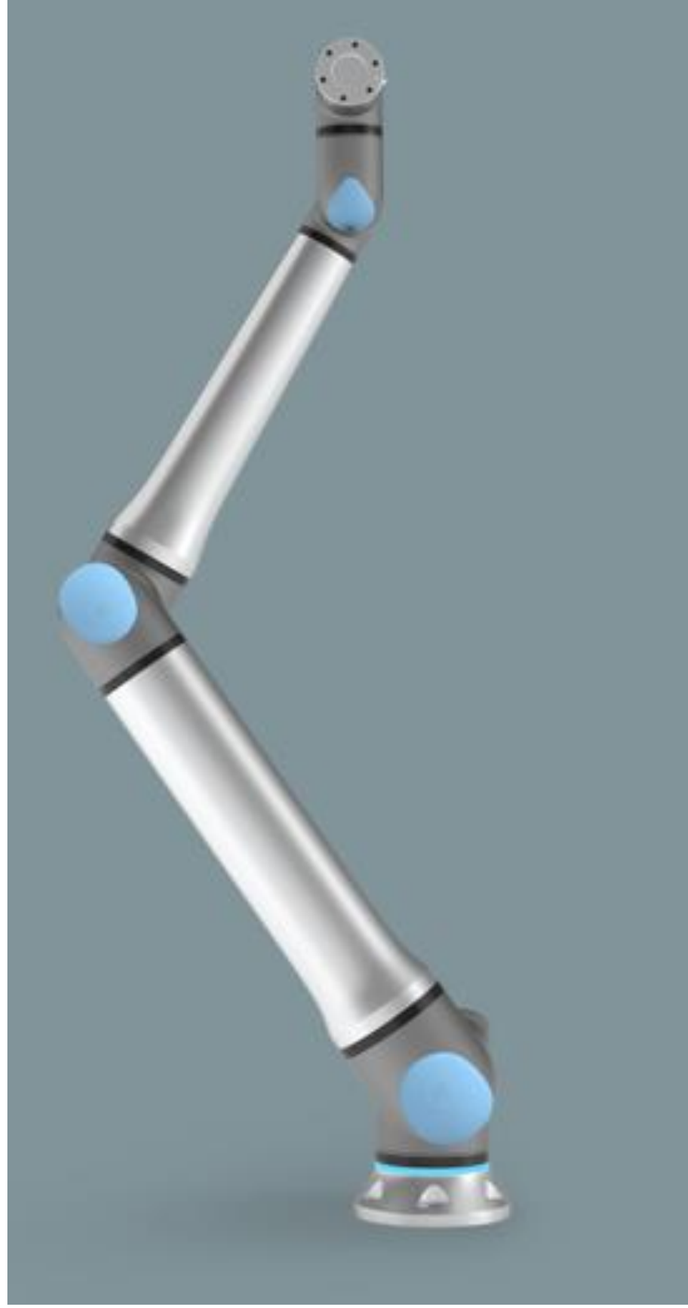
Protection from Environment



IP RATING

- UR20/UR30 Robots are IP65 rated.
- E-series Robots are IP54 rated.
- This refers to how well protected they are from solids/dust and liquids in the environment:

IP	5	4
"Ingress Protection"	First Digit: Solids Protection	Second Digit: Liquids Protection

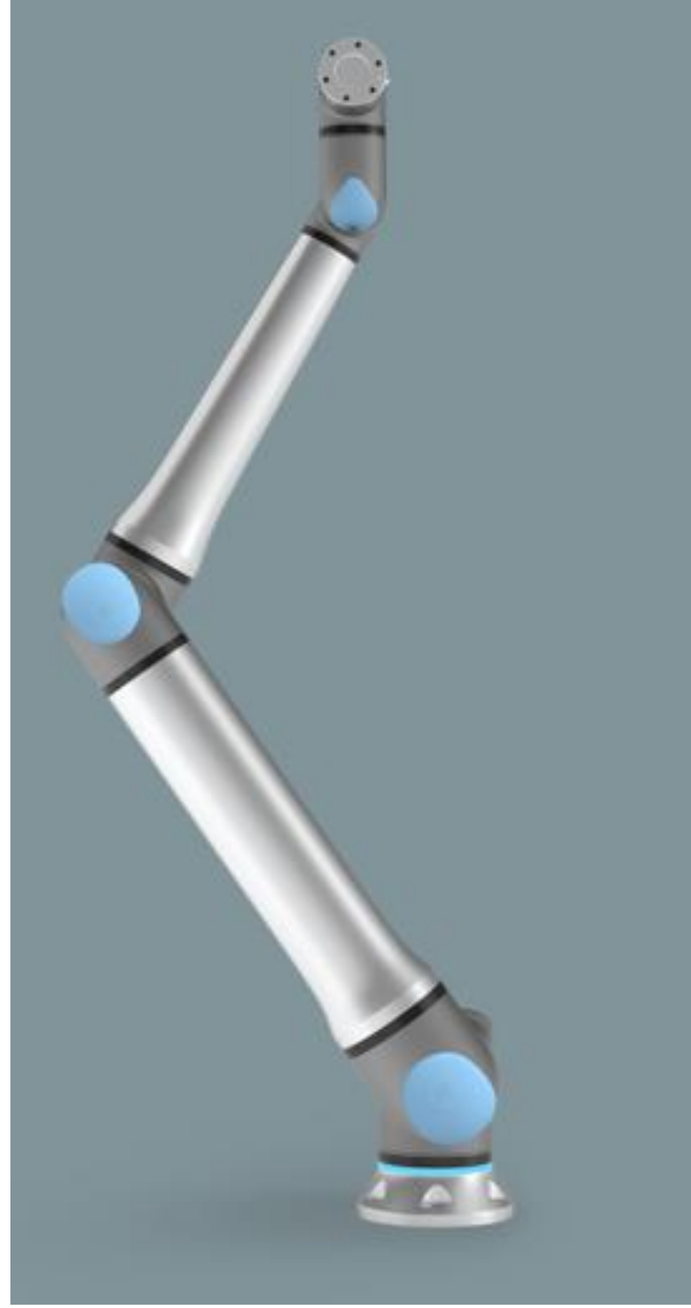


UNIVERSAL ROBOTS

IP RATING - SOLID

- IP54 for solids means that small dust particles may be able to penetrate the robot seals.
- It is therefore important to keep the outer surface of the robot free from dust during regular maintenance.

Level	Object size protected against	Effective against
0	Not protected	No protection against contact and ingress of objects
1	>50mm	Any large surface of the body, such as the back of the hand, but no protection against deliberate contact with a body part.
2	>12.5mm	Fingers or similar objects.
3	>2.5mm	Tools, thick wires, etc.
4	>1mm	Most wires, screws, etc.
5	Dust Protected	Ingress of dust is not entirely prevented, but it must not enter in sufficient quantity to interfere with the satisfactory operation of the equipment; complete protection against contact.
6	Dust Tight	No ingress of dust; complete protection against contact.

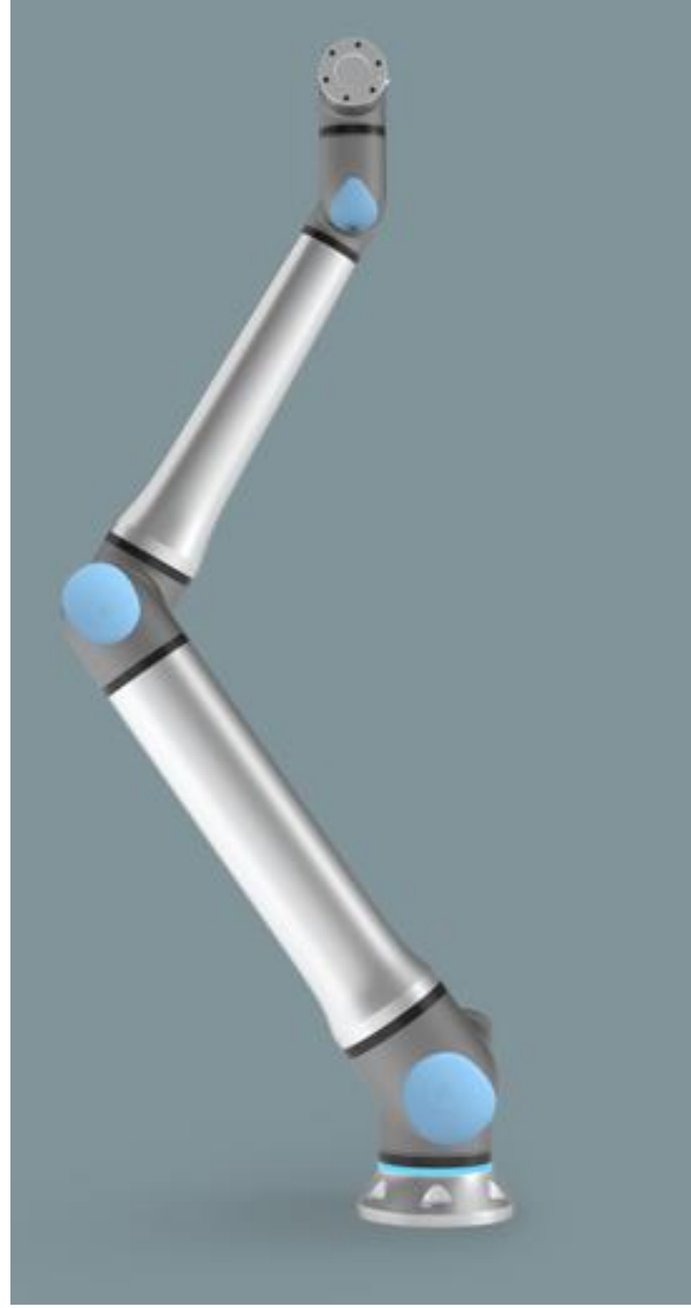


UNIVERSAL ROBOTS

IP RATING - LIQUID

- IP54 for liquids means that any water dripping or splashing on the robot should not affect operation
- Water projected at the robot under pressure may get inside the robot and should therefore be avoided.

Level	Object size protected against	Effective against
0	Not protected	–
1	Dripping water	Dripping water (vertically falling drops) shall have no harmfull effect.
2	Dripping water when tilted up to 15°	Vertically dripping water shall have no harmful effect when the enclosure is tilted at an angle up to 15° from its normal position.
3	Spraying water	Water falling as a spray at any angle up to 60° from the vertical shall have no harmful effect.
4	Splashing water	Water splashing against the enclosure from any direction shall have no harmful effect.
5	Water jets	Water projected by a nozzle (6.3mm) against enclosure from any direction shall have no harmful effects.



UNIVERSAL ROBOTS

CORROSIVE FLUIDS

- The IP rating for liquids on the previous slides only applies to water and other similar non corrosive fluids.

- If a cutting fluid or coolants common in CNC applications meet the robot they may degrade the rubber seals, decreasing the IP rating of the robot.

- If this type of fluid is likely to contact the robot, a protective suit should be considered



UNIVERSAL ROBOTS

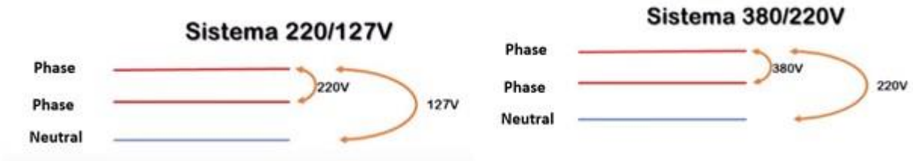
PROTECTIVE SUITS

- If the robot is to be operating in an environment where the IP rating may be insufficient, a protective suit is a good countermeasure.
- There are numerous UR+ certified protective suits available through the UR+ showroom:
- US/EU Brands: Roboworld, Robo-Gear, EVOTEC
- Asian Brands: Shanghai Chunyu, Shenzhen Tongxing



ROBOT -> ELECTRICAL GRID

- Recall: control box mains connection specs are 100-240VAC, 50-60Hz, Single Phase
- Before connecting robot to any plug in your factory, use a stable voltage line OR connect a stabilizer and/or Universal Power Supply
- If connecting robot to panel and NOT electrical grid, ensure that it has 100-240VAC power to the robot controller, and has fuse/circuit breaker
- Ensure robot receives SINGLE PHASE as its input

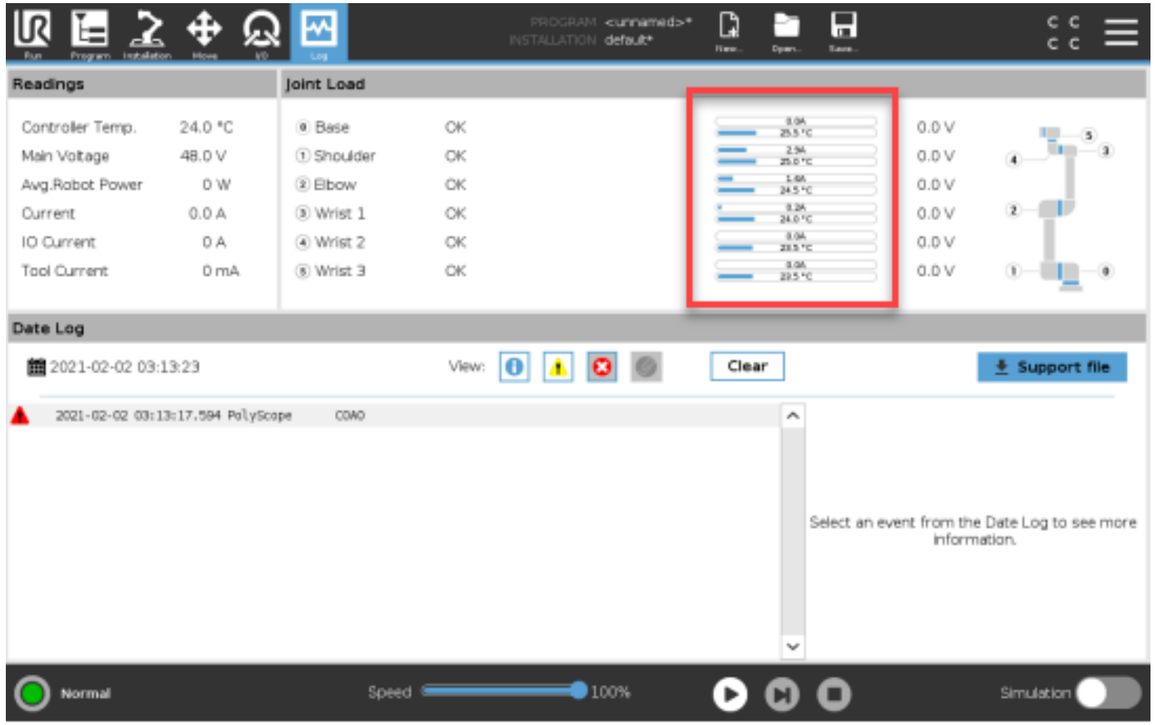


- Always ensure that the ground pin of the robot controller socket or electrical power panel is connected to earth
- You CAN use an external PS (24V) to feed digital IO terminals inside the control panel…need to remove jumper & supply + & - from power supply to control panel

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COLD ENVIRONMENTS

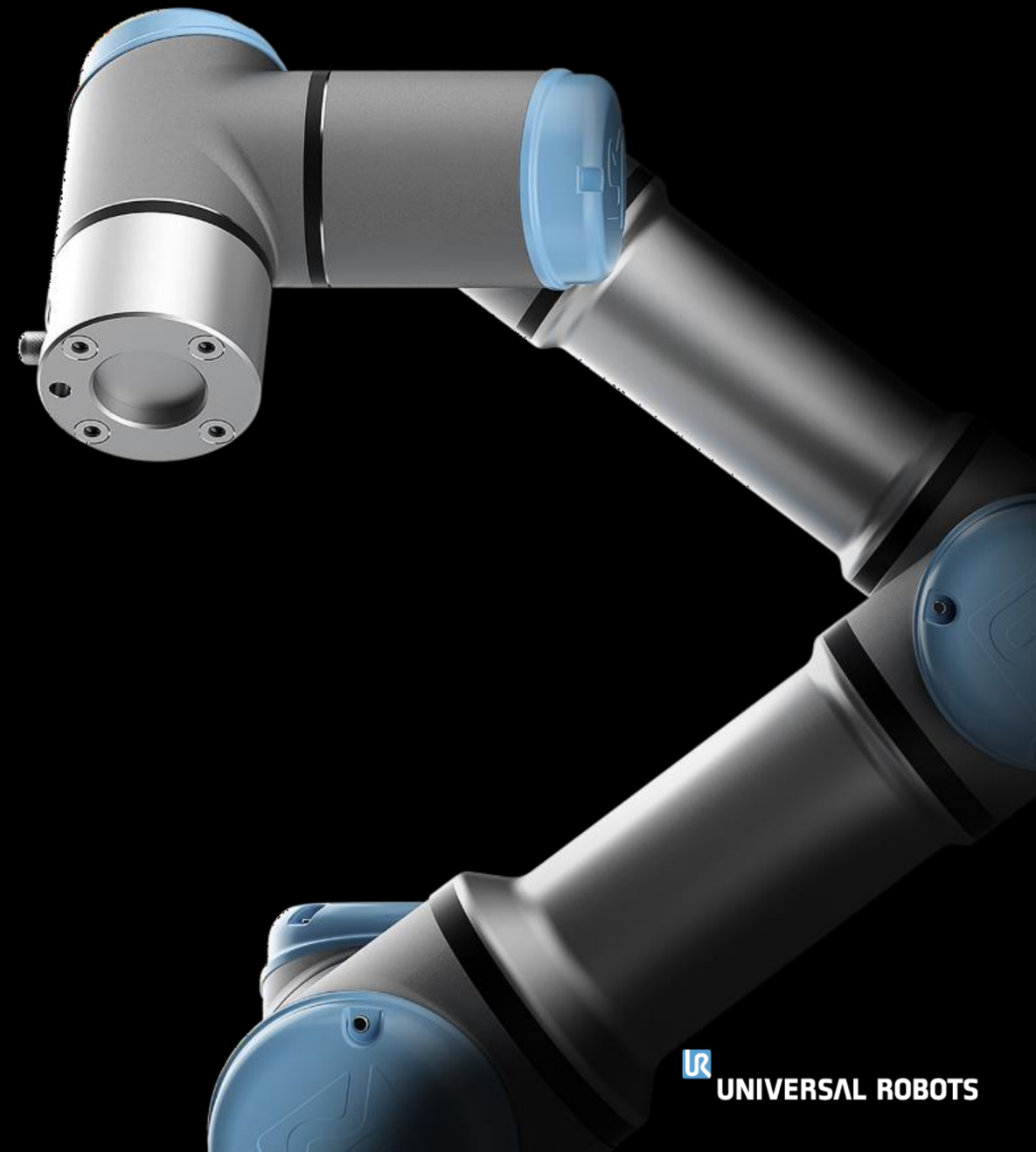
- If the robot is to be operating in an environment where the ambient temperature is very close to 0°C, may need to incorporate warmup routine
- This is particularly relevant if robot is sitting, then starts up after a long period in said cold environment
- Not recommended to operate robot at high speeds until joints have reached internal operating temps
- Solution: do slow movements until robot joints reach internal operating temperature before reaching higher speeds.



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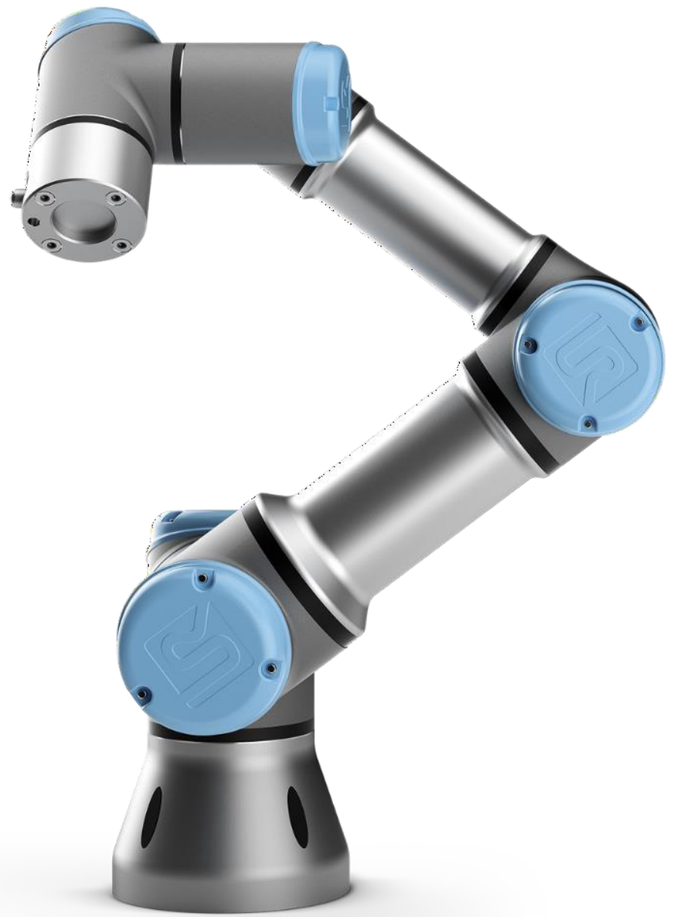
INSTALLATION

Payload Configuration



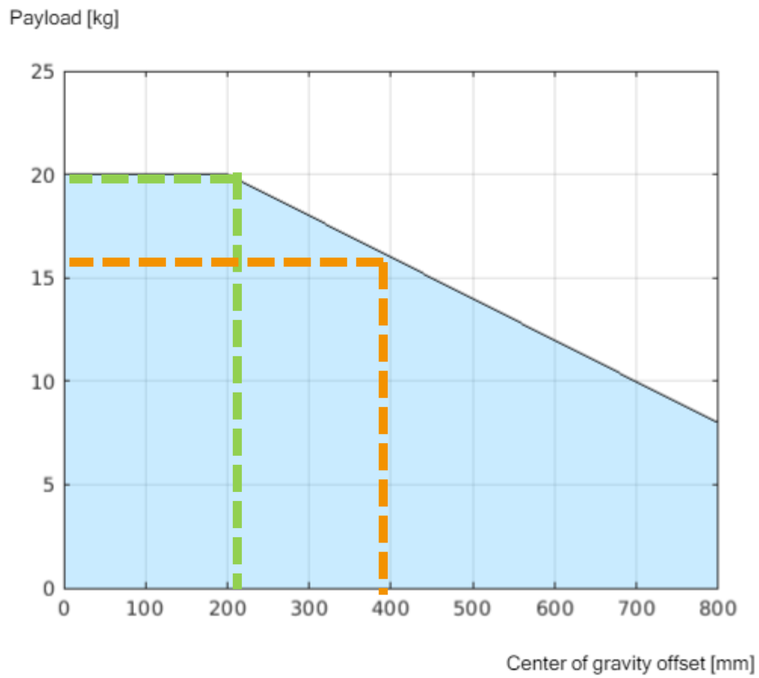
PAYLOAD CONFIGURATION

- Payload Mass and Center of Gravity settings in Polyscope must reflect actual payload at all times.
- Initial settings are configured under Installation TCP Setup tab or under the Payload tab for users running Polyscope 5.10+. This is sufficient for applications such as dispensing, screwdriving or polishing where the payload remains constant.
- For applications involving transferring of products, payload must also be updated during using script command when payload changes. A Set Payload command exists for users running Polyscope 5.10+
- Incorrect payload settings can lead to inaccurate/unstable robot movements, frequent protective stops interrupting robot operation and potentially shorten the working life of the robot.



PAYLOAD CONFIGURATION

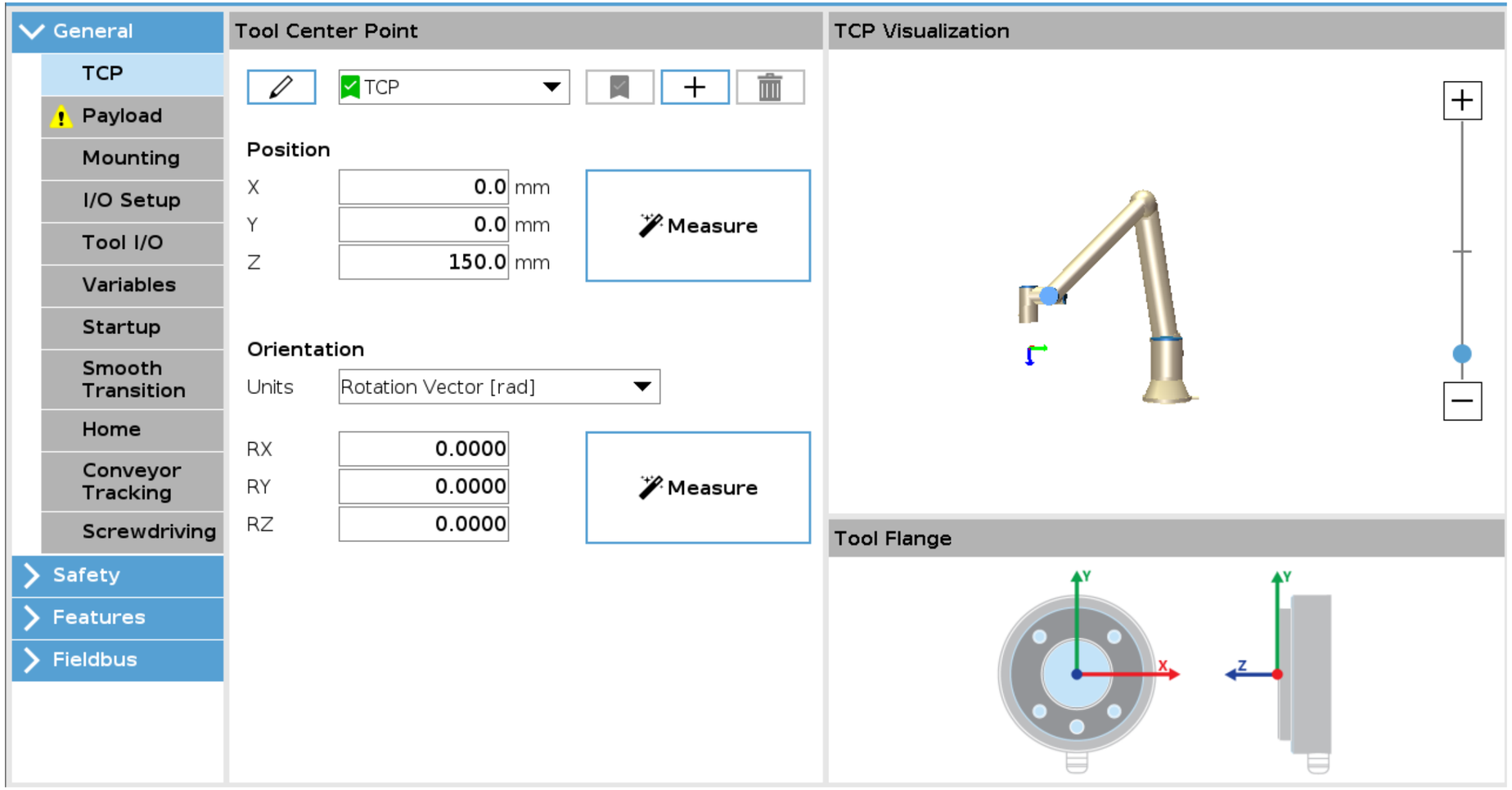
- As the centre of gravity moves further away from the robot tool the maximum allowed payload weight decreases.
- Each UR user manual shows the relationship for that robot model. The UR20 graph is shown on the right.
- UR20 can support full 20kg payload up to 200mm offset. A payload with 400mm CoG offset should not exceed 16kg in weight.
- Exceeding maximum rated payload may affect robot lifetime and warranty coverage.



The relationship between the rated payload and the center of gravity offset.

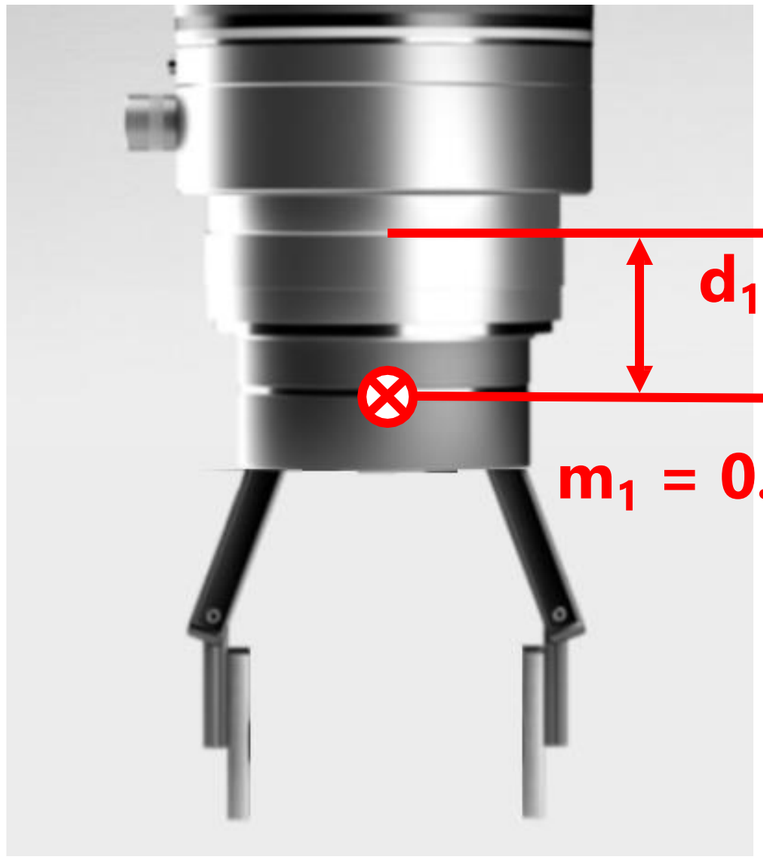
PAYLOAD CONFIGURATION

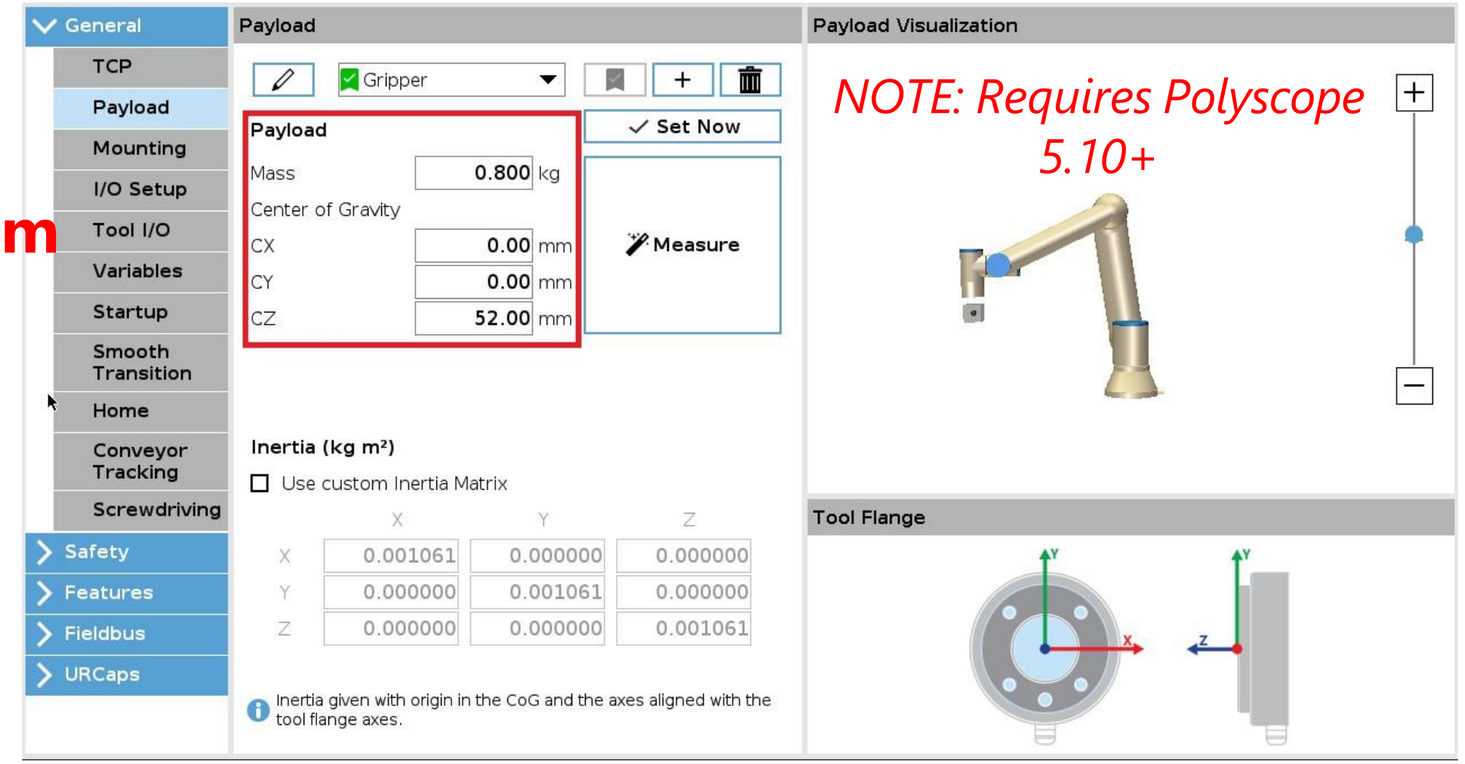
- Pick and place example: Confirm TCP position/orientation of empty gripper and set this in Installation TCP tab:



PAYLOAD CONFIGURATION

Add in the mass and CoG and name the Payload (optionally the user can hit "set now" to set it as the active payload)

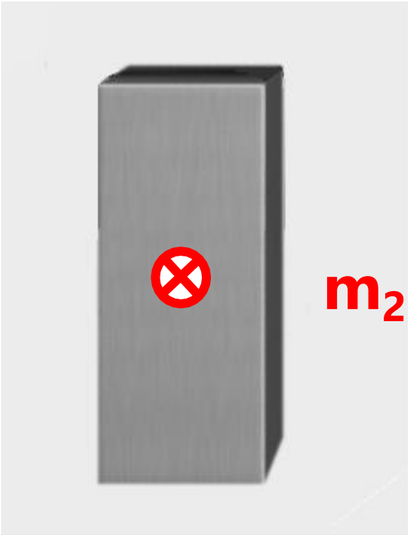




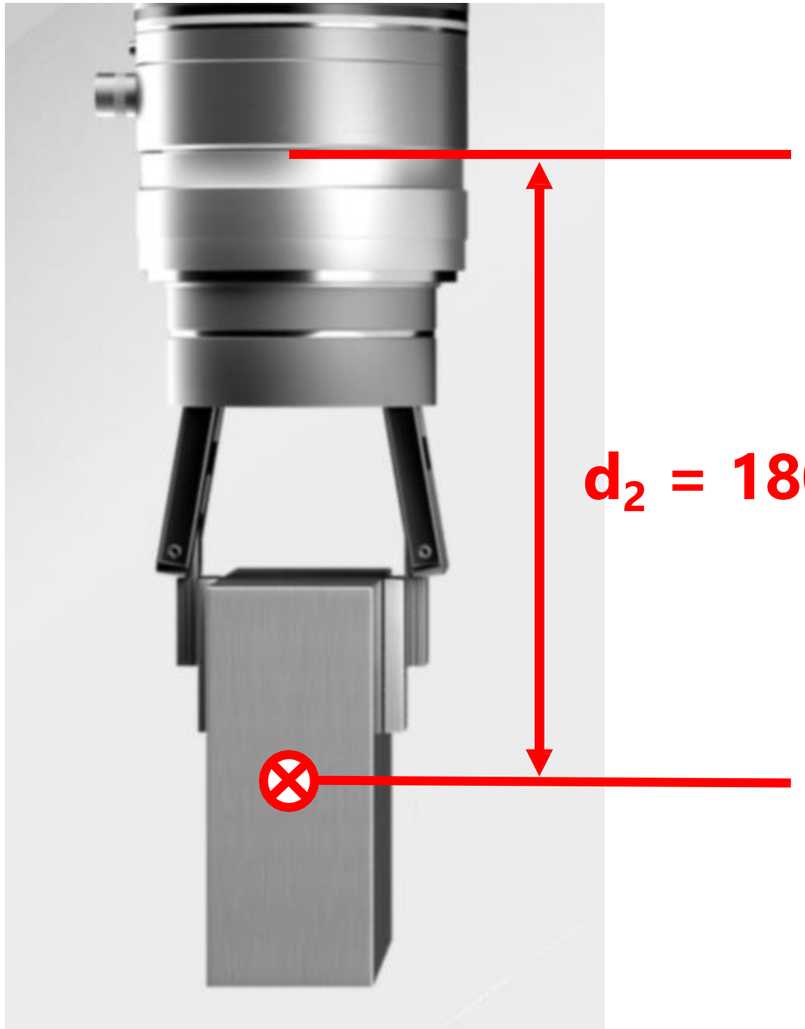
UNIVERSAL ROBOTS

PAYLOAD CONFIGURATION

- Confirm weight of product and CoG distance when held by gripper.



m_2 = 1.35kg



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PAYLOAD CONFIGURATION

- Calculate the combined CoG offset of the gripper

$$\frac{m_1 d_1 + m_2 d_2}{m_1 + m_2} = \frac{0.80kg \times 0.052m + 1.35kg \times 0.180m}{0.80kg + 1.35kg} = 0.132m$$

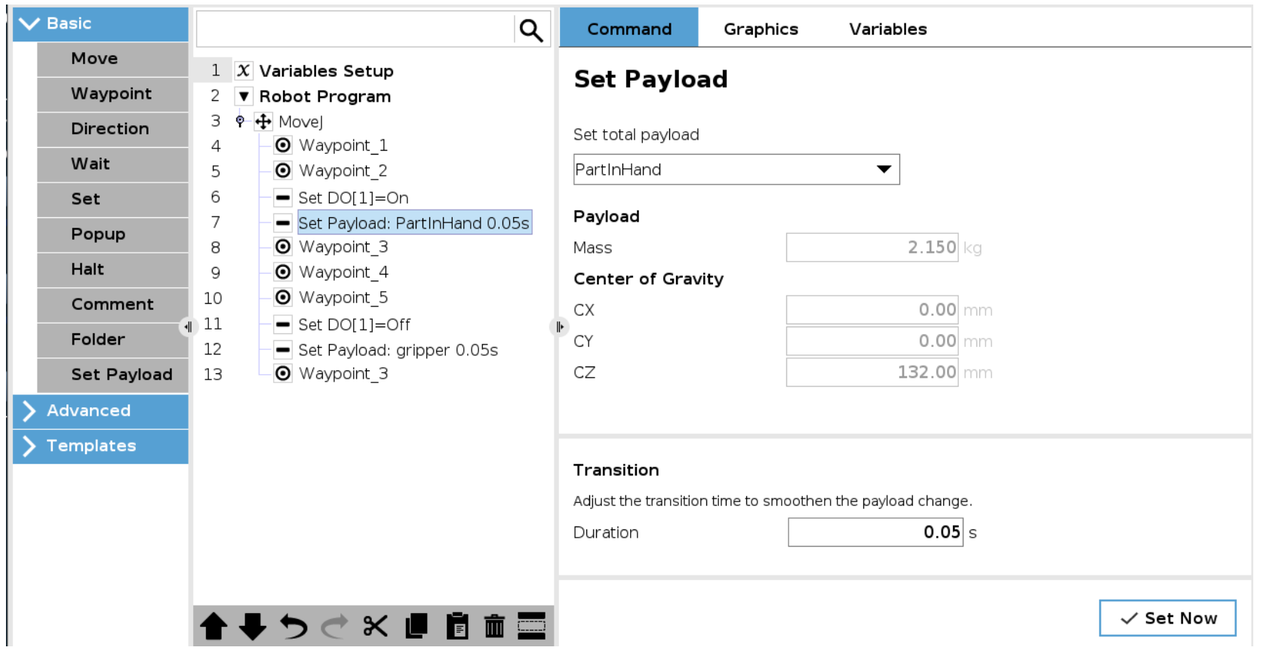
- When picking/placing update using script command in program:

- set_target_payload(mass, CoG, inertia, transition_time)
- mass = payload weight in kilograms (kg)
- CoG = a list of three positions [Cx, Cy, Cz] in meters (m)
- Inertia: payload inertia matrix (in kg*m^2), as a vector with six elements [Ixx, Iyy, Izz, Ixy, Ixz, Iyz] with origin in the CoG and the axes aligned with the tool flange axes.
- Transition_time: the duration of the payload property changes in seconds

- We only have Cz offset in this example, so Cx and Cy = 0

PAYLOAD CONFIGURATION

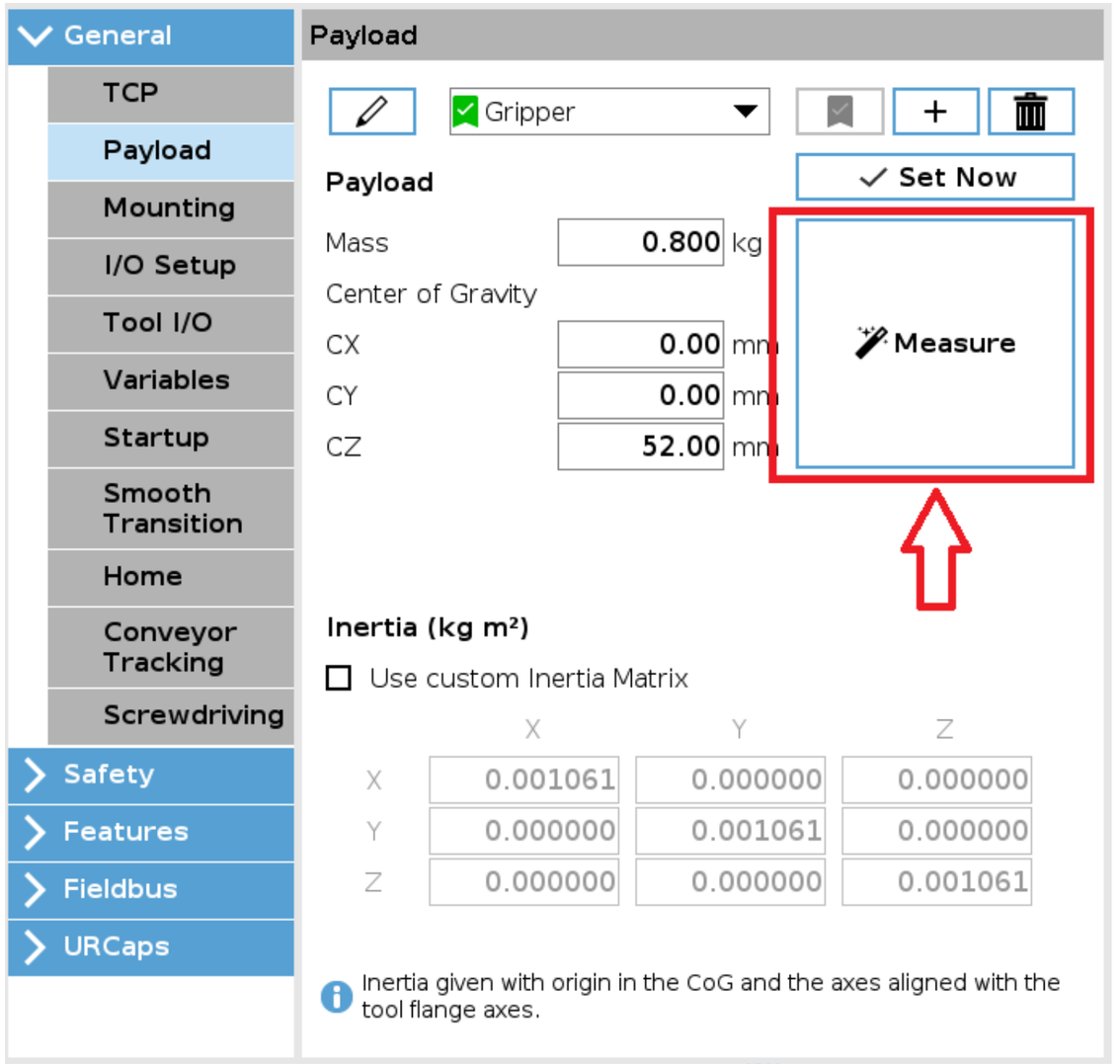
- User should create an additional Payload (in this case named "PartInHand")
- No need for additional script commands in Polyscope 5.10+ as all payload and CoG information is taken care of in the "Set Payload" command at the bottom of the Basic tab once configured in the Installation tab



UNIVERSAL ROBOTS

PAYLOAD ESTIMATION

- In E-series & newer robots, users can utilize a wizard to allow the robot to estimate the payload and CoG of its end effector utilizing the built-in FT sensor
- In the Installation tab, navigate to General-> Payload and click on the Measure icon
- Follow the steps in the wizard and record the robot in three(3) or four(4) unique positions
- Once completed, the robot will provide an estimated payload and CoG. Adjust CoG values as necessary



UNIVERSAL ROBOTS

PAYLOAD CONFIGURATION

- When working with multiple grippers on the robot at the same time it is even more important to adjust payload and CoG accurately.

- For more complex payload offsets the method listed on previous slides can be used to calculate for different axes.

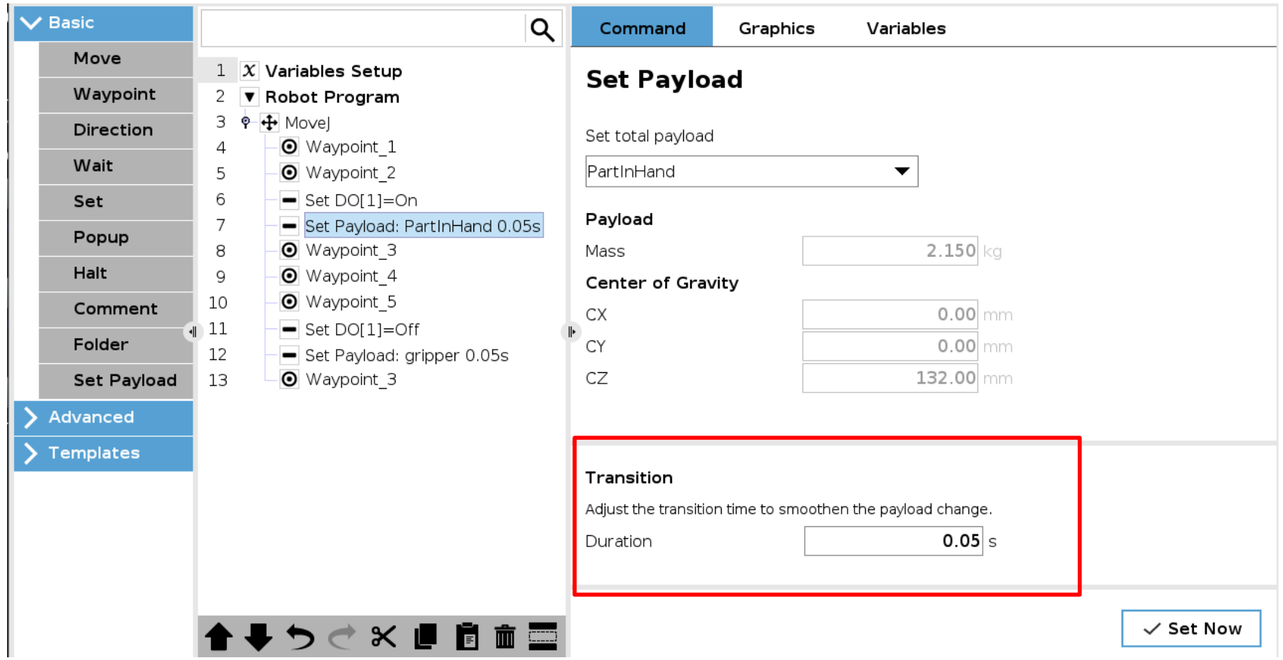
- CAD packages such as Solidworks can be used to give the overall centre of mass of end effector with different payloads. This is a convenient approach if all model data is available.



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PAYLOAD TRANSITION TIME

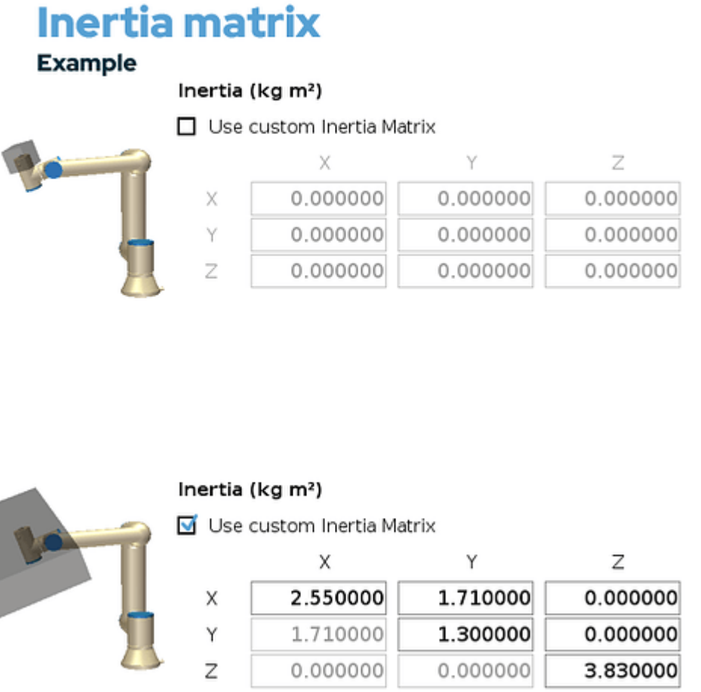
- This is the time it takes the robot to adjust for a given payload
- You can add a payload transition time in seconds.
- Setting a transition time larger than zero, prevents the robot from doing a small "jump",when the payload changes. The program continues while the adjustment is taking place
- Using the Payload Transition Time is recommended when picking up or releasing heavy objects or using a vacuum gripper



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INERTIA MATRIX

- If your payload is long or has large surface area, using inertia matrix is essential (think of a tool in image below)
- Inertia matrix indicates how the mass of the payload is distributed
- In the Installation tab, navigate to General-> Payload and click on the ‘use custom inertia matrix’ box
- Most CAD programs can calculate the inertia matrix for you, i.e. SolidWorks (units in kg m^2)

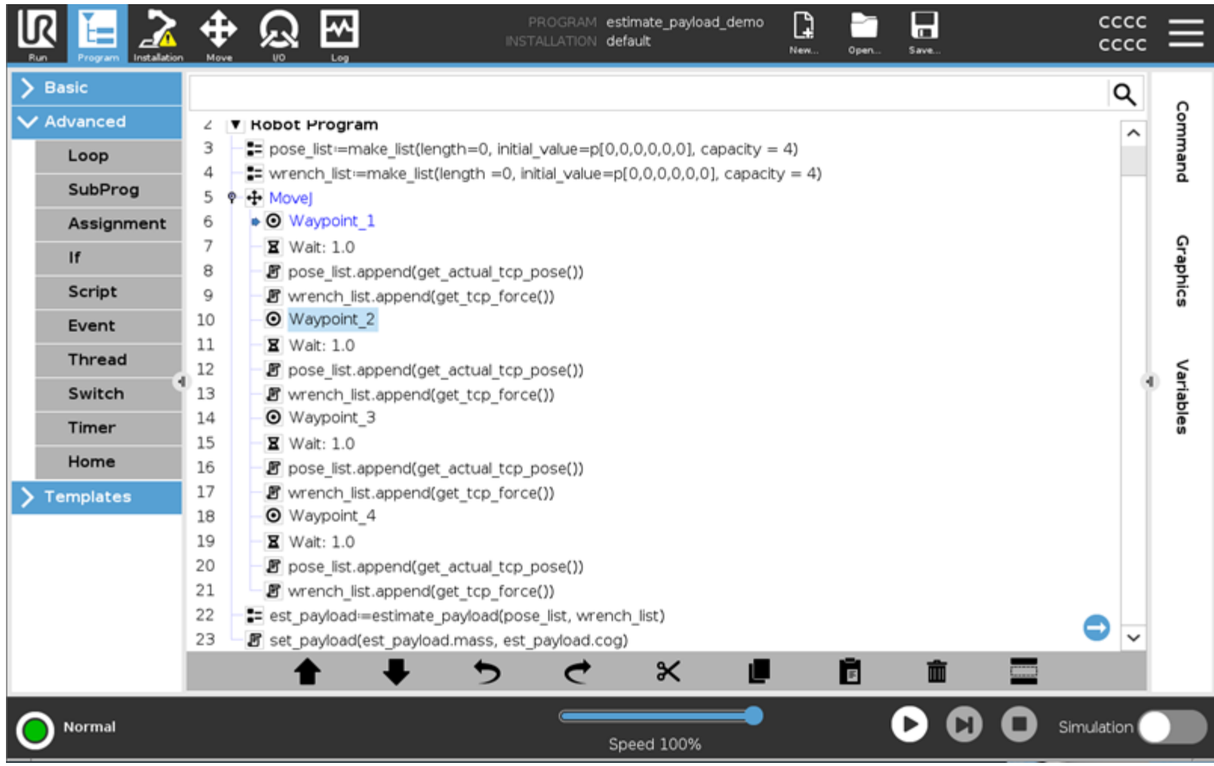




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ESTIMATE PAYLOAD

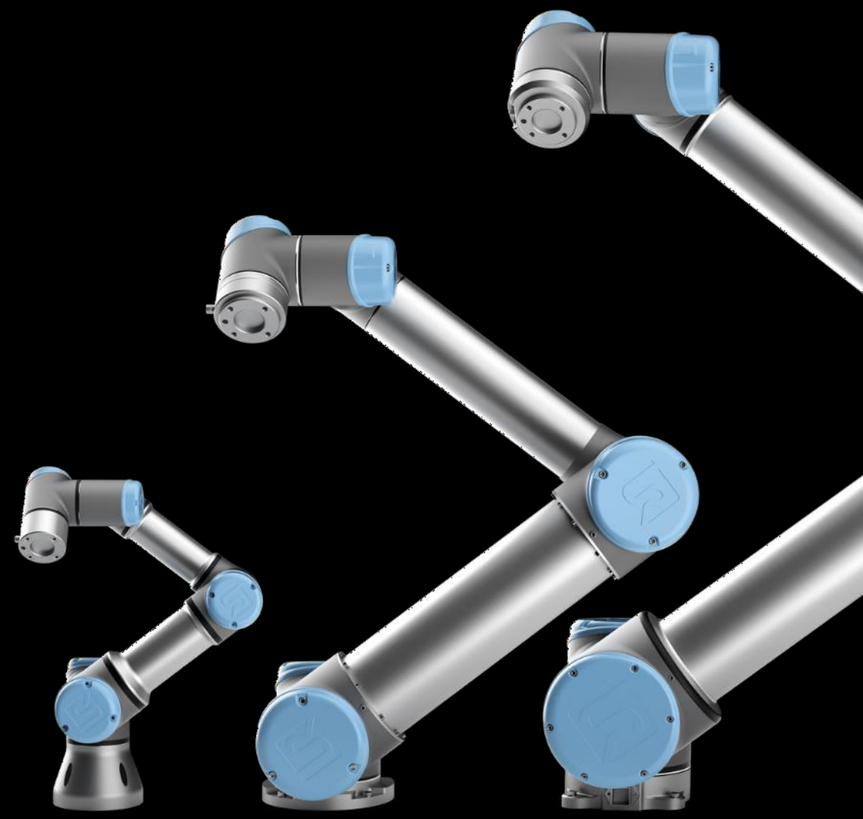
- There exists a script function estimate_payload() which can be used to estimate payload during program execution
- Allows the payload mass and center of gravity to be estimated inside a robot program, based on a list of poses and force torque recordings
- A way to use script to set payload similar to how the wizard does it



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MOTION PROGRAMMING

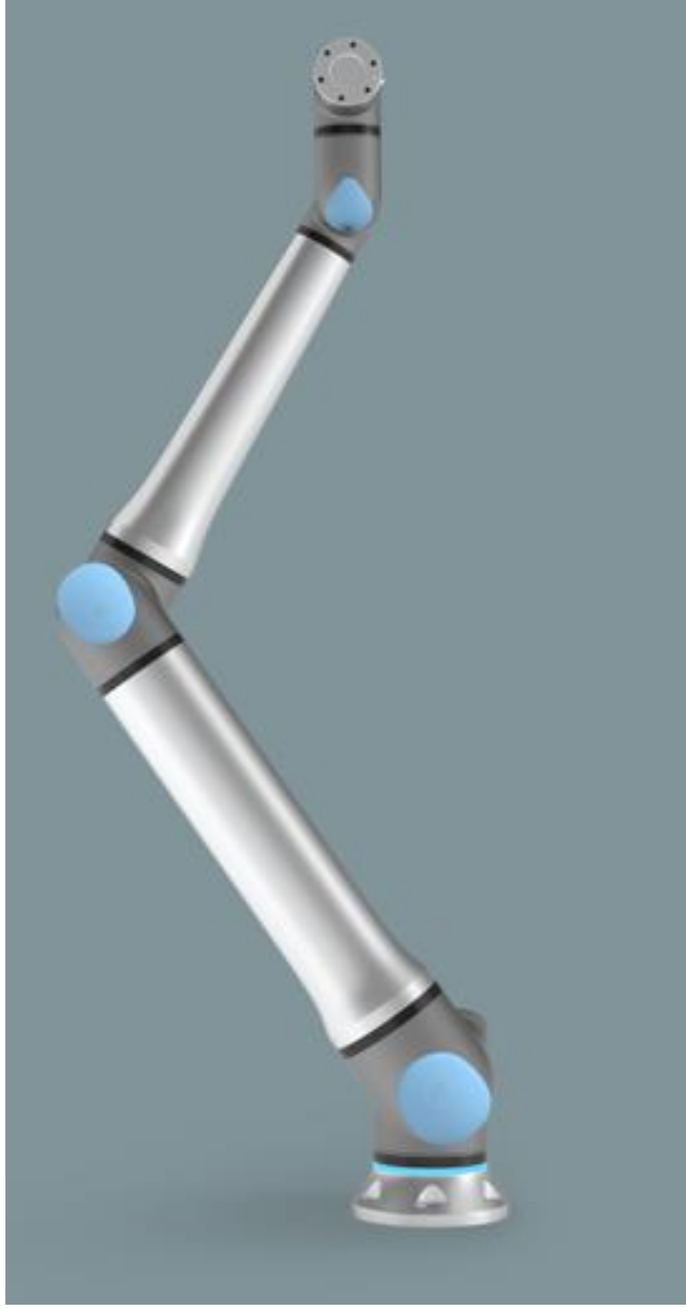
Blend Radii



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BLEND RADII

- A blend radius allows the robot to move more efficiently through paths where it doesn't need to stop and complete an action.
- Instead of decelerating to a complete stop at a waypoint and then accelerating again after it, a blend allows the robot to move close to a waypoint without stopping, saving time.
- This results in shorter cycle times without needing to increase speed or acceleration of motions.
- All paths should be optimized with blend radii before increasing speed or acceleration.



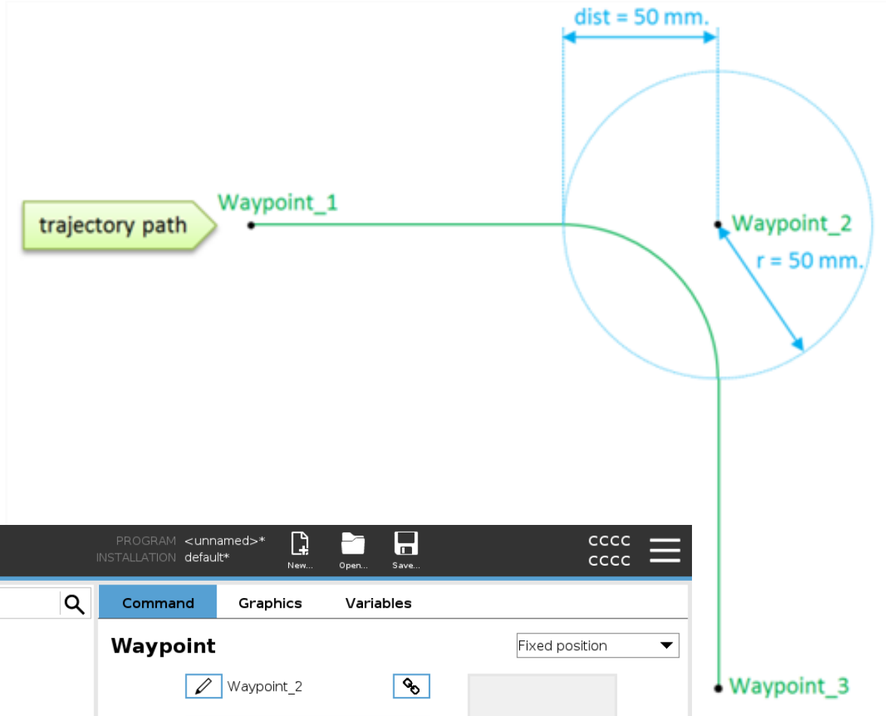
UNIVERSAL ROBOTS

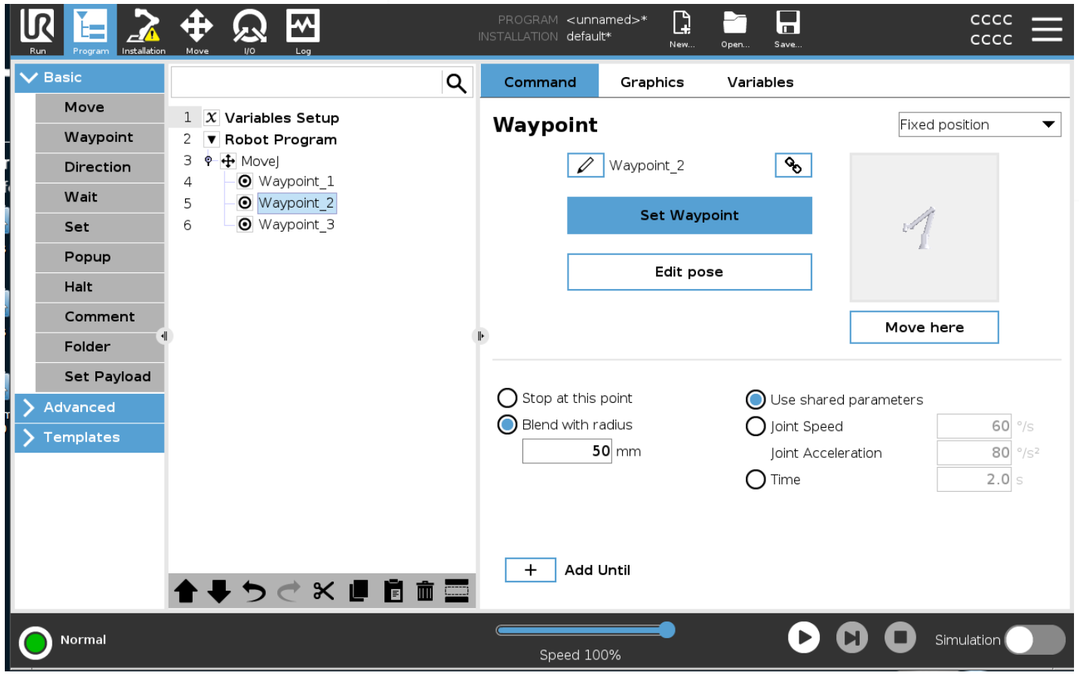
BLEND RADII

- A blend radius can be added to a MoveL or MoveJ to optimize the path.

- A smaller value will make the robot move closer to the waypoint while a larger value will allow for a smoother path. (Minimum blend size generally shouldn't be below 15mm.)

- MoveP uses blends to maintain a constant speed throughout a path for process applications such as dispensing. MoveP should not be used unless this constant speed is specifically required for the process.

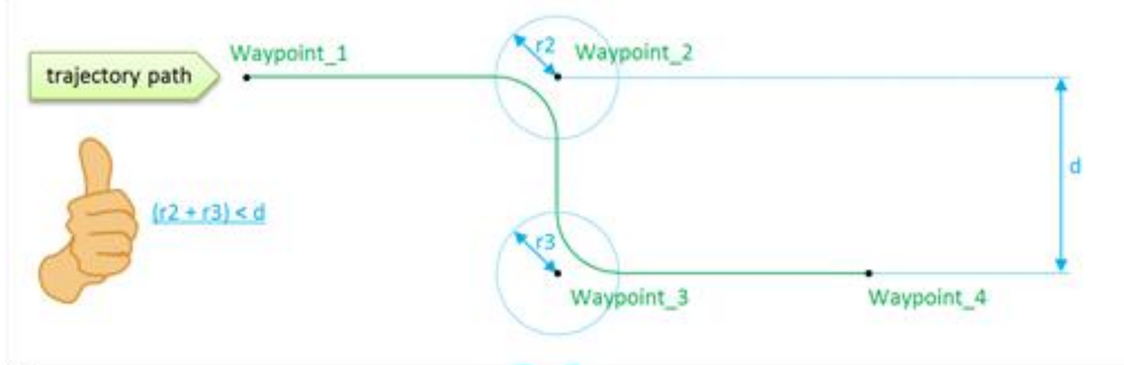


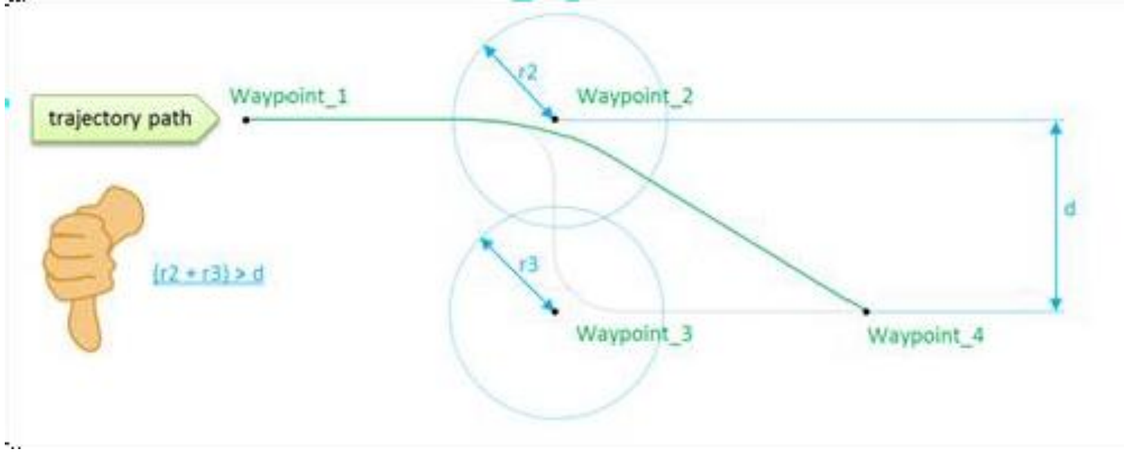


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OVERLAPPING BLENDS

- Blends must be far enough apart that they do not overlap.
- Overlapping blends cause the waypoint to be skipped, resulting in unintended robot motion paths.
- Polyscope will not allow user to enter blend values that will lead to overlapping but in certain circumstances it is still possible.
- If there are overlapping blend warnings in the log file, the blends radius must be reduced.







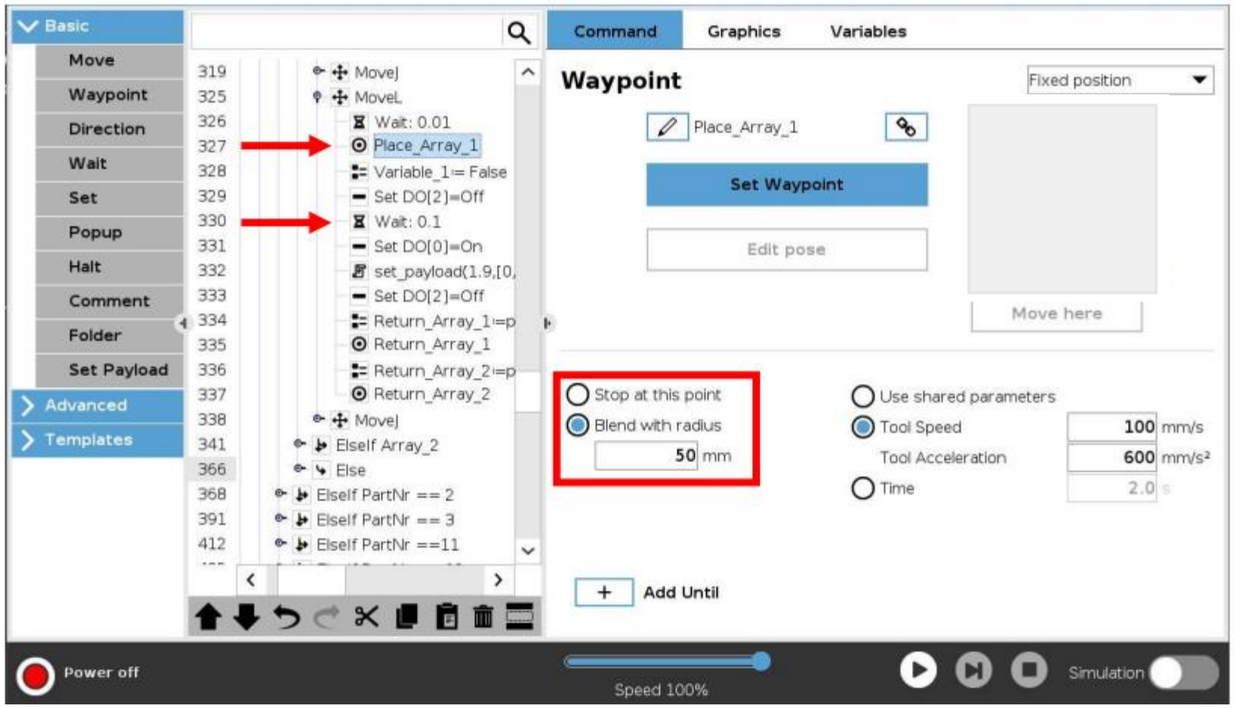
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OTHER BLEND CONSIDERATIONS

- Aborted/Incomplete blend-if there are blend continuity issues, can cause unintended effects

- Aborted blend example-waypoint is configured with a blend followed by a blocking function-can cause unnecessary hard decelerations
    - Blocking means the caller waits until the callee finishes its processing

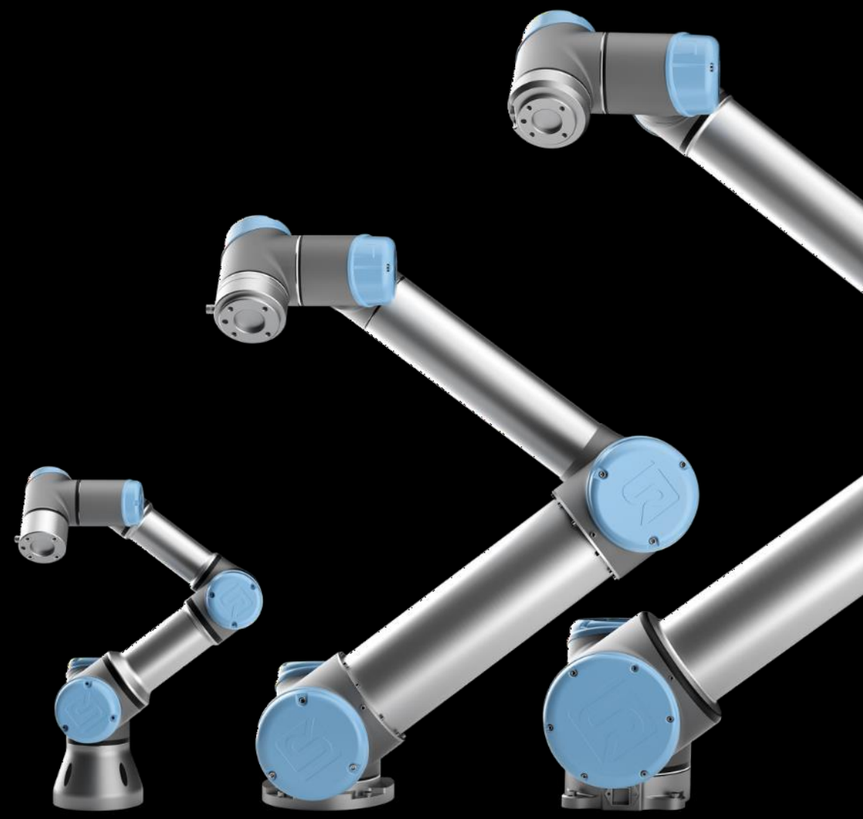
- Small blends-small blend radii can make the path turn sharper, and oftentimes this can lead to jerky motion paths



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MOTION PROGRAMMING

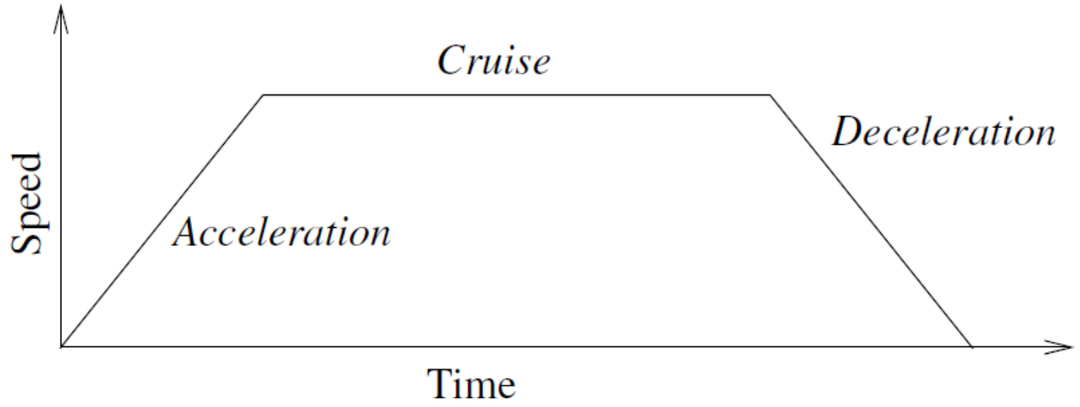
Acceleration Values



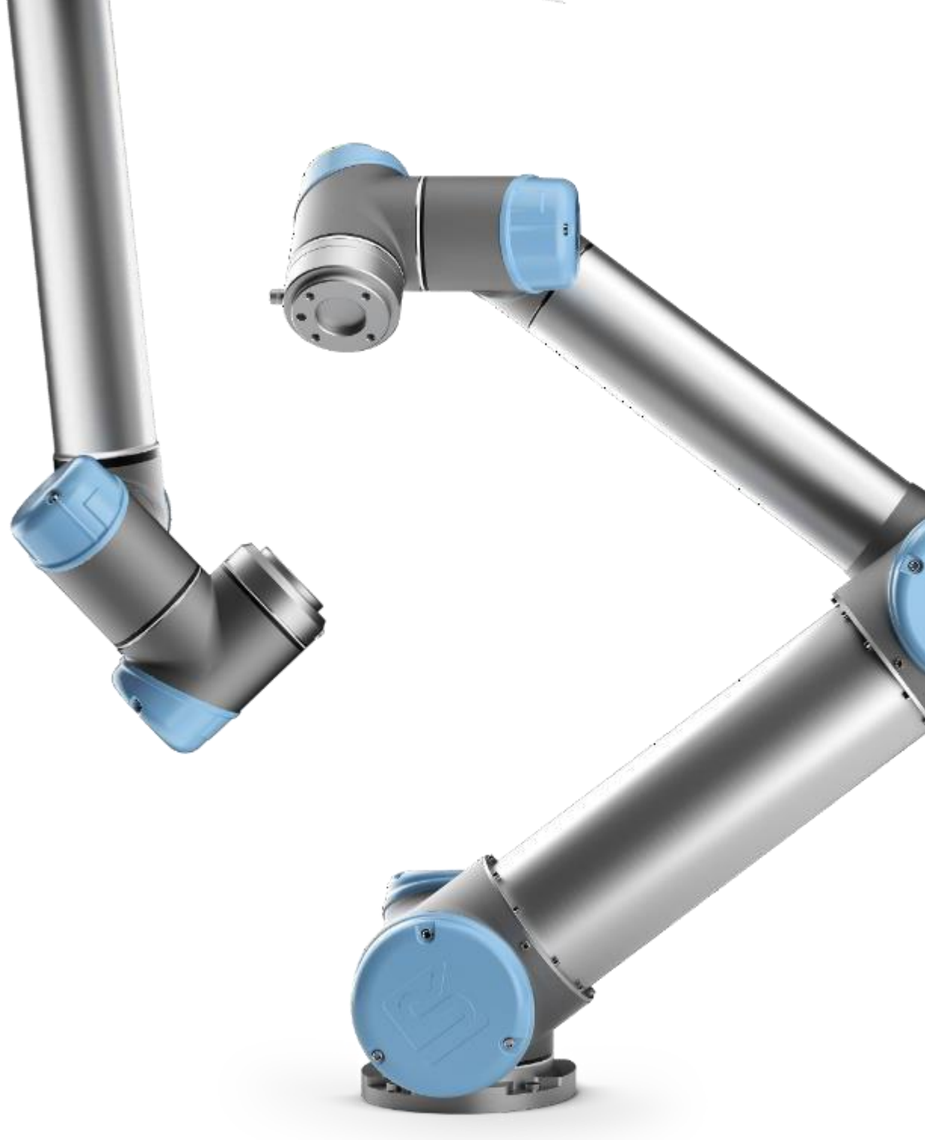
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ACCELERATION VALUES

- The acceleration parameter for a move command determines how quickly the robot reaches the target speed for that movement.



- Acceleration values should be set as low as possible while still achieving cycle time requirements.



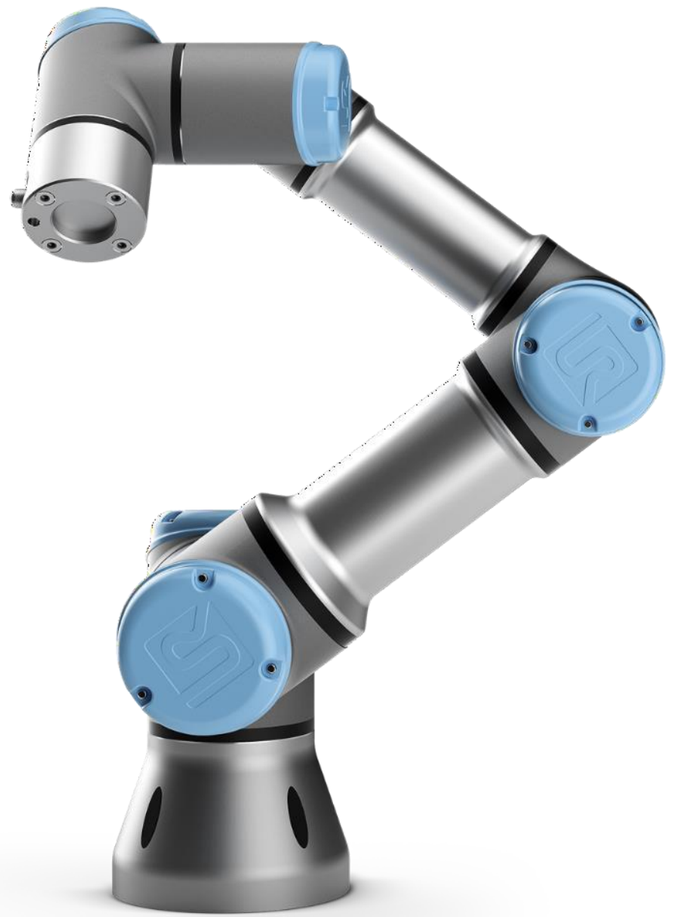
ACCELERATION VALUES

- Paths, blends and wait positions should all be optimised before the user increases acceleration to meet cycle time requirements.

- Higher acceleration requires higher torque and therefore puts more stress on the gearboxes in robot joints.

- Maximum recommended values are:
  - MoveJ: 800°/s^2
  - MoveL: 2500mm/s^2

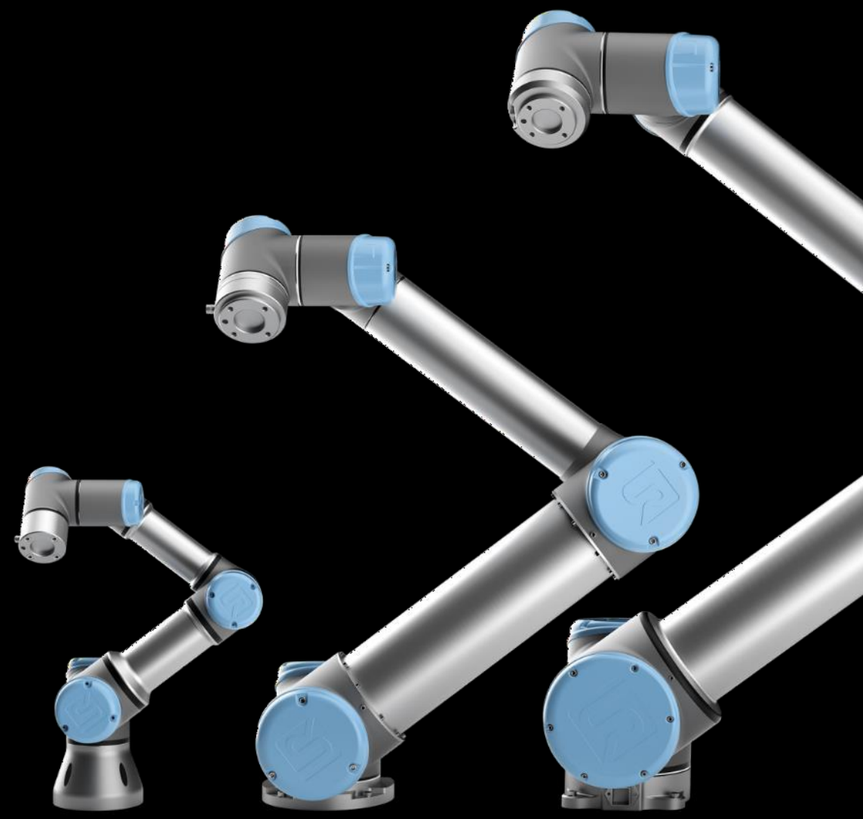
- Exceeding these values may reduce lifetime of robot joints and affect warranty coverage.



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MOTION PROGRAMMING

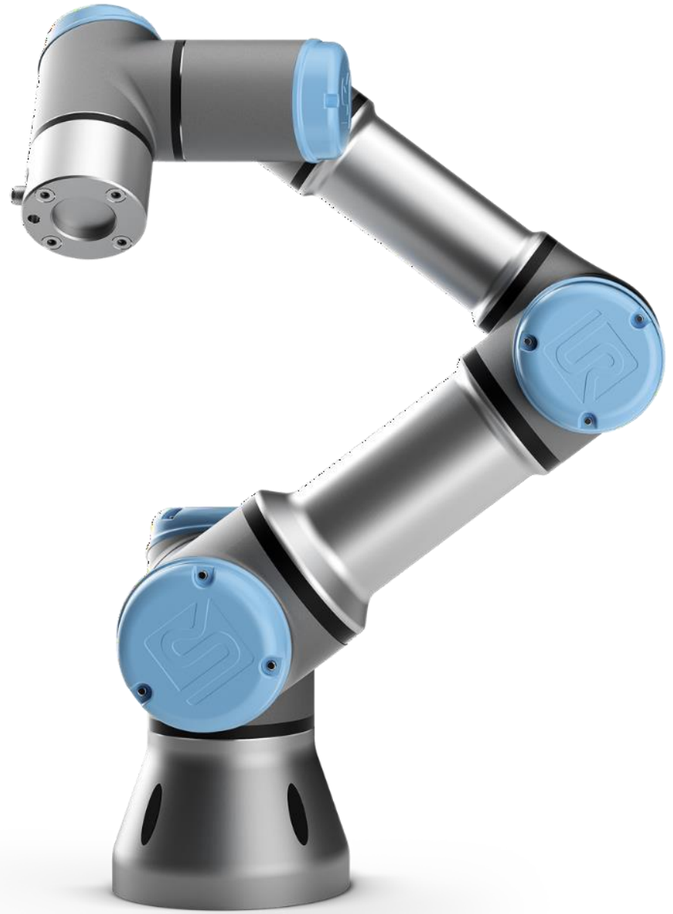
Singularities



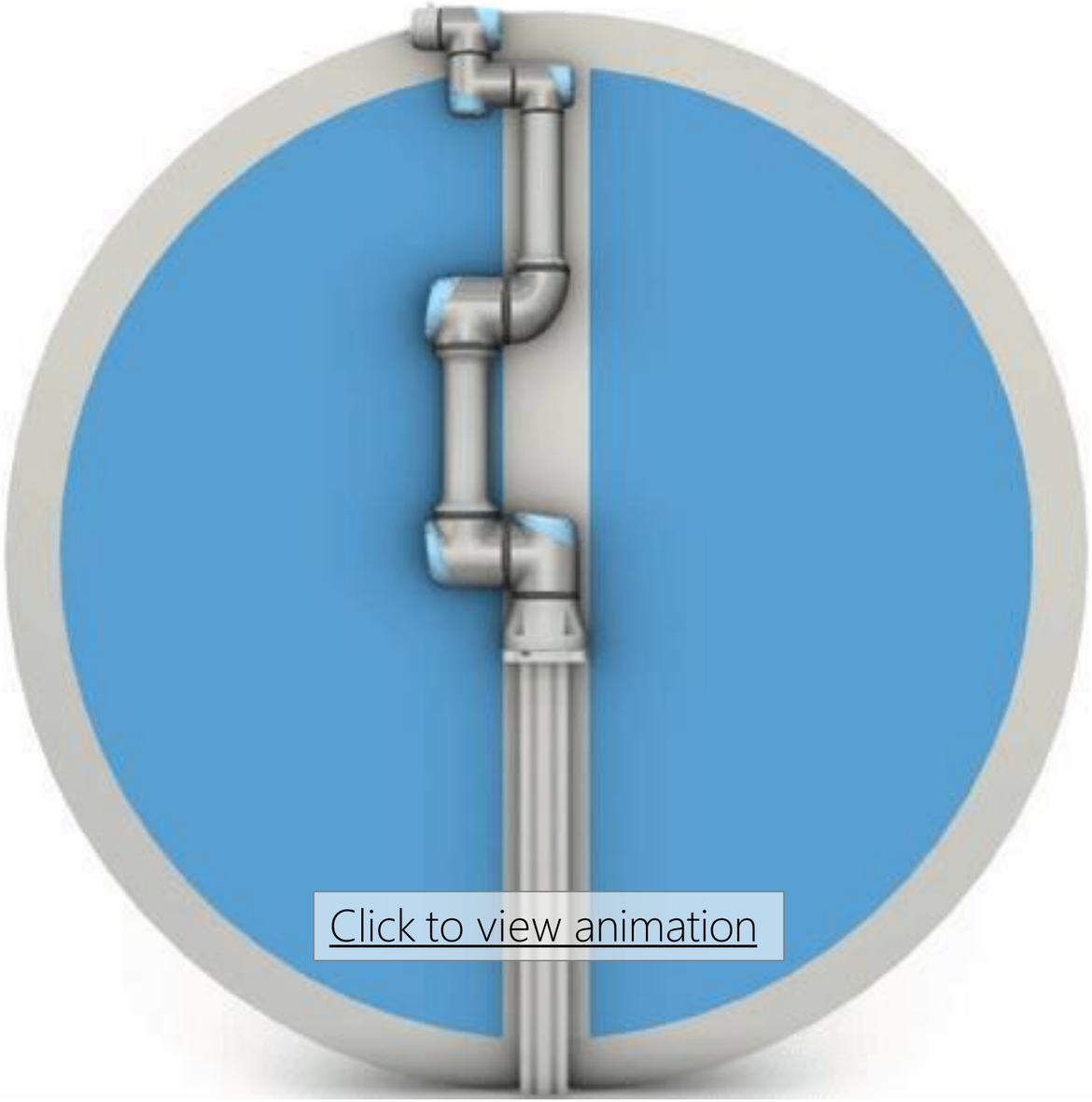
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SINGULARITIES

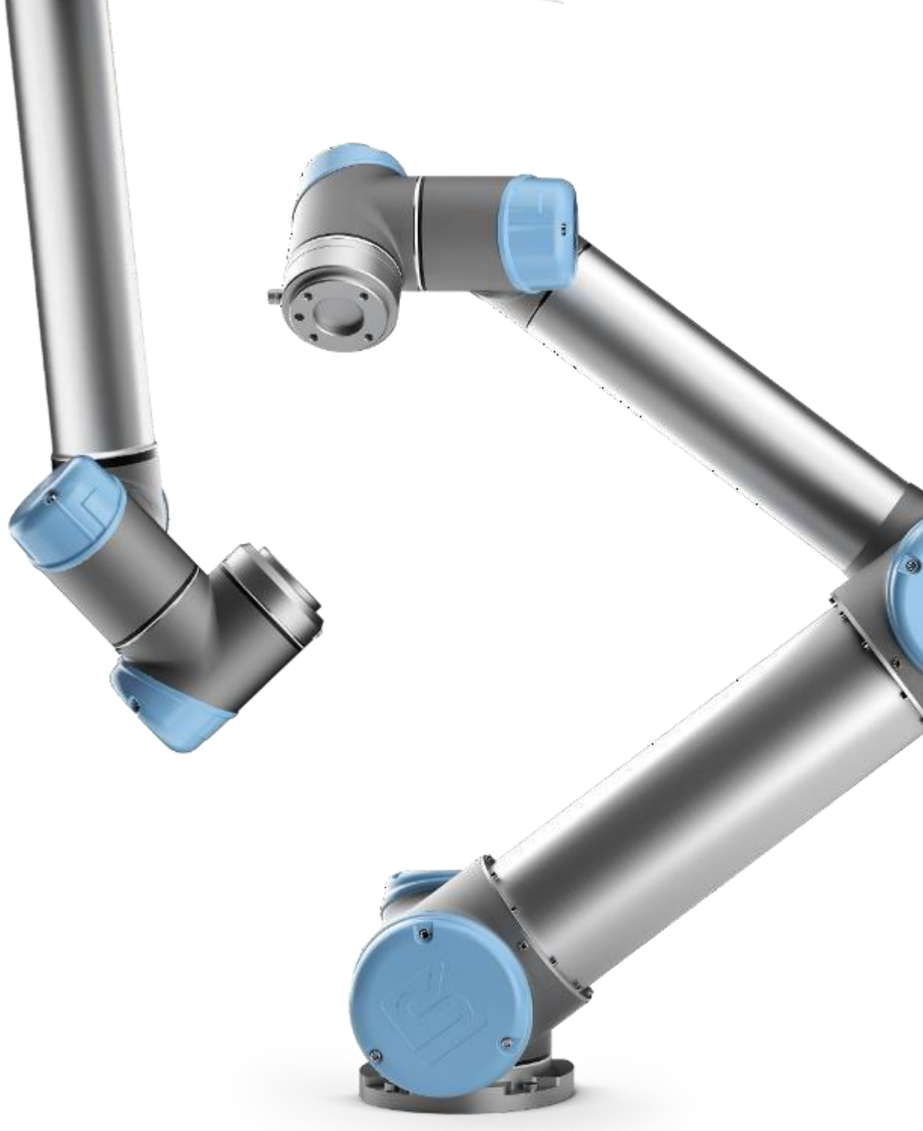
- A singularity is a scenario in which a set of joint positions or speeds which allow the robot to move its TCP in the desired way cannot be found.
- There are three main singularity conditions on any UR robot as explained in the following slides.
- Moving too close to a singularity can result in unexpected increased in joint speed and protective stops.
- Users must avoid programming the robot to operate close to these singularity conditions, as the sharp movements that can occur as a result raise concerns around safety and robot joint lifetime.



OUTER WORKSPACE SINGULARITY



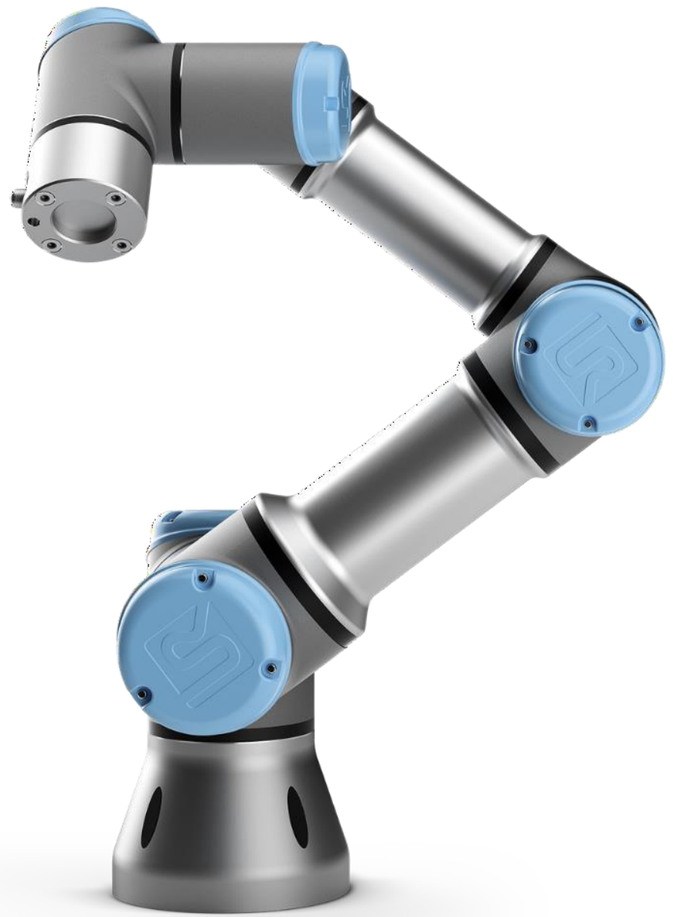
Click to view animation



UNIVERSAL ROBOTS

INNER WORKSPACE SINGULARITY



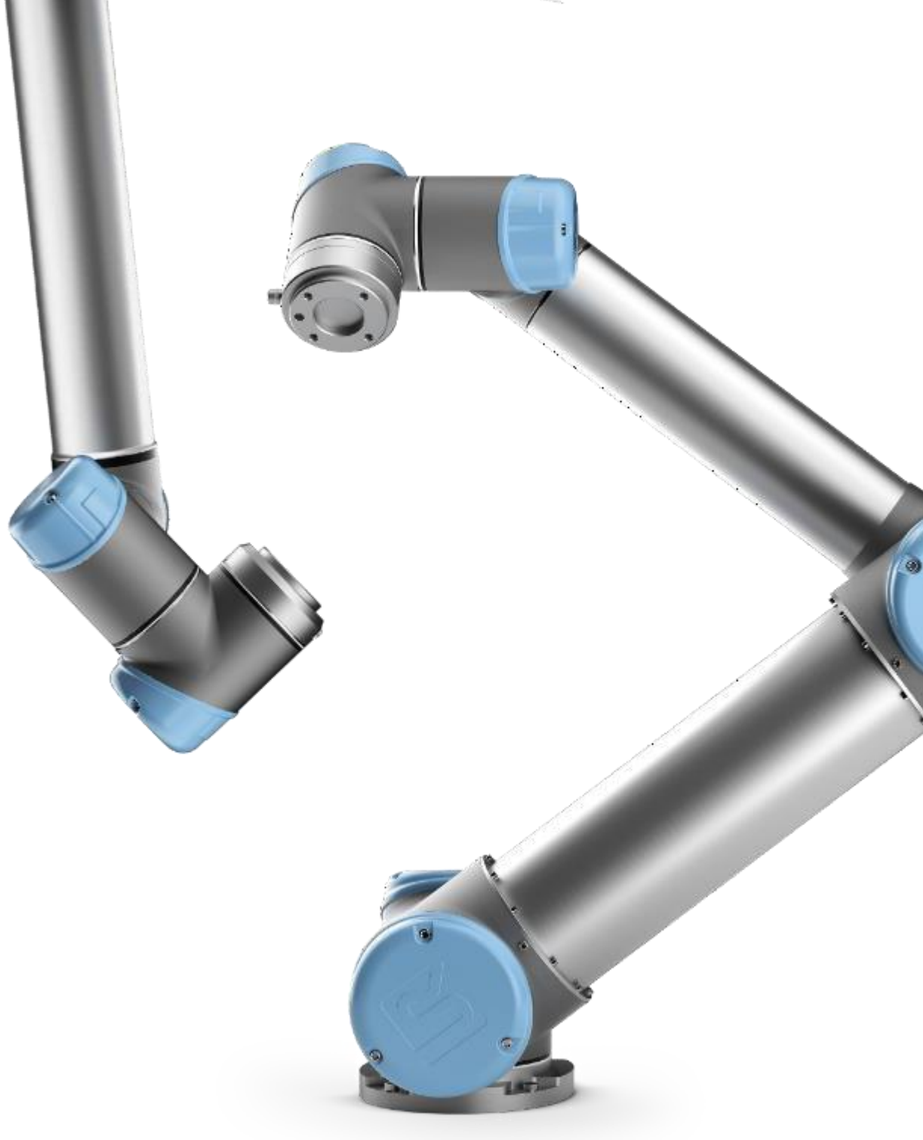


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WRIST ALIGNMENT SINGULARITY



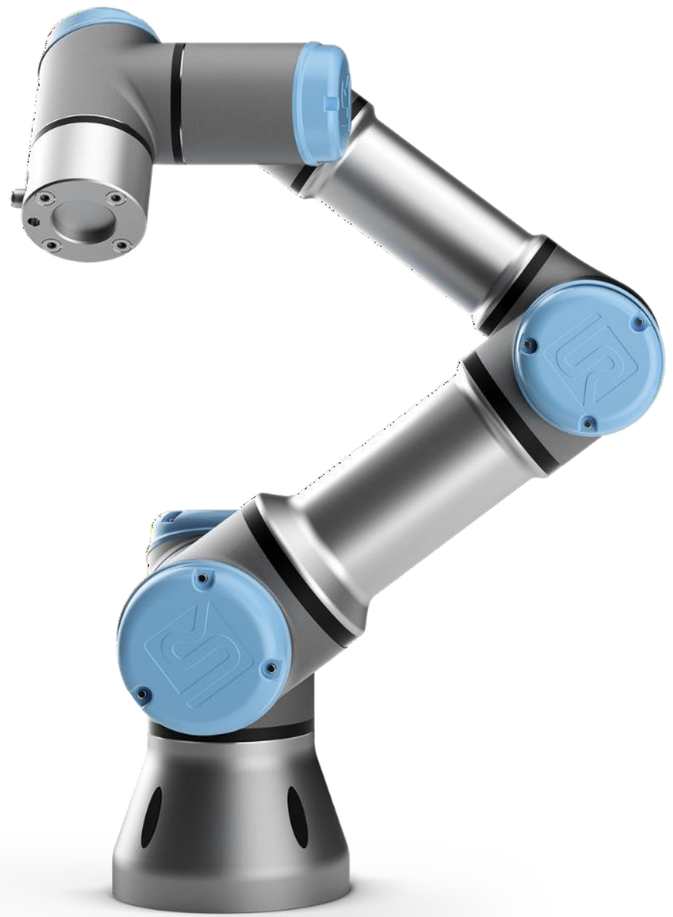
Click to view animation



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SINGULARITIES

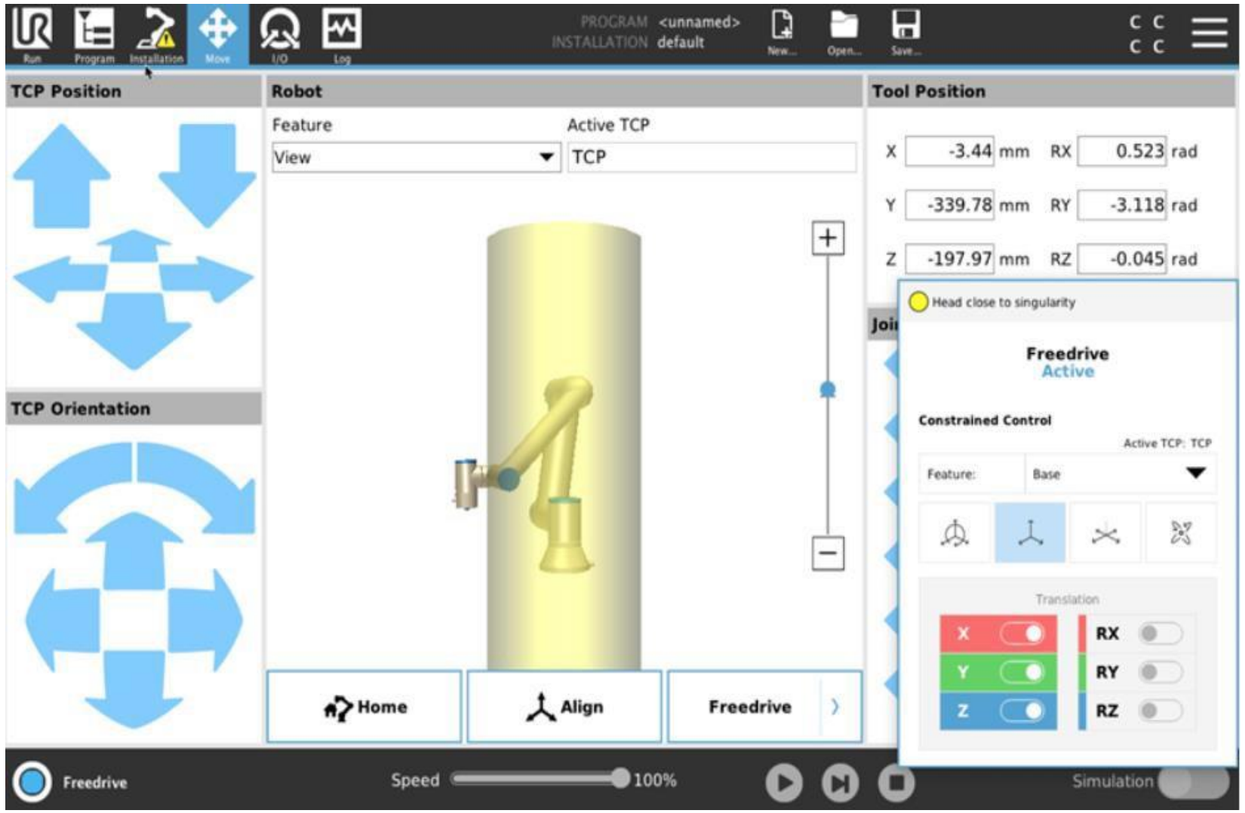
- If a singularity is affecting the robots motions, one of the following actions could resolve the problem:
- Change the move type from MoveL to MoveJ
- Move the robot base or surrounding equipment so the robot is not operating in this singularity zone.
- Change orientation or length of tooling to change the joint configuration required to reach a desired TCP position.

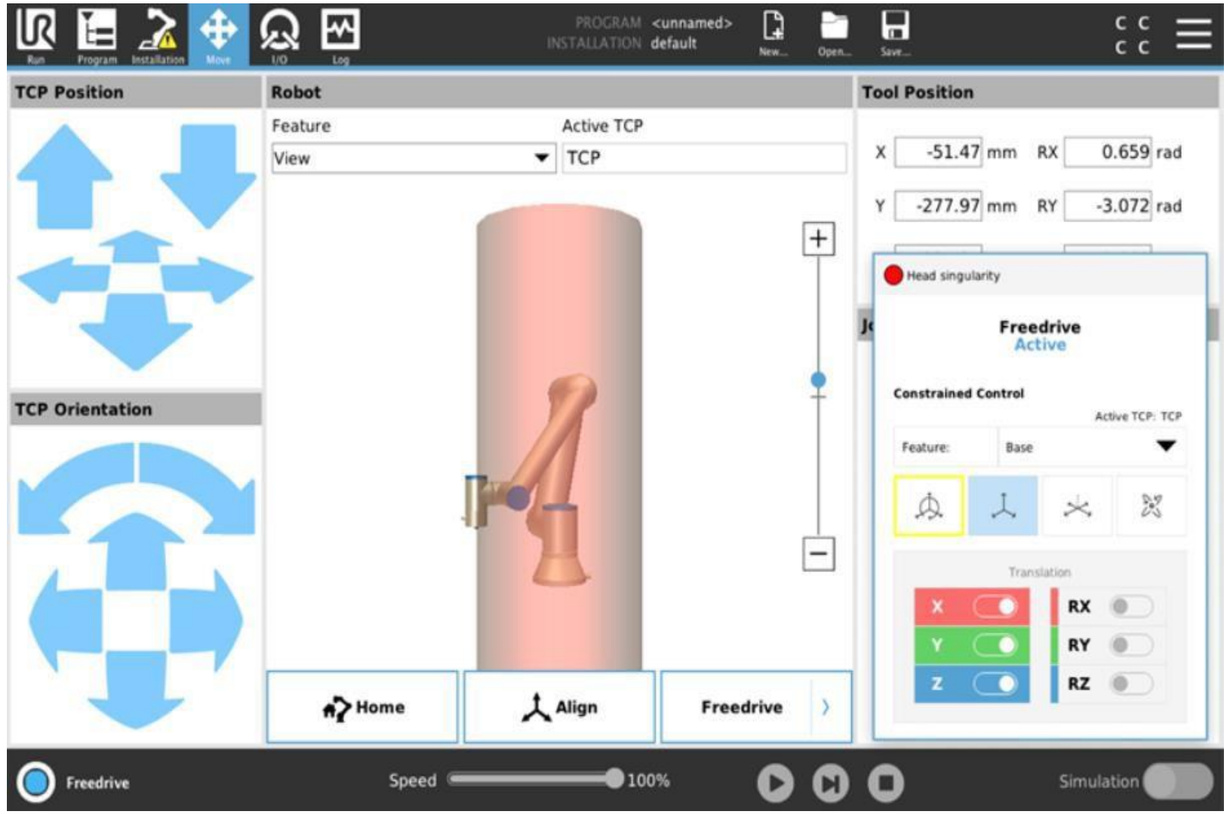


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SINGULARITIES

- In Polyscope 5.10+, the Move tab will indicate when the robot is approaching or in a singularity as well as highlighting which joint(s) are affected





UNIVERSAL ROBOTS

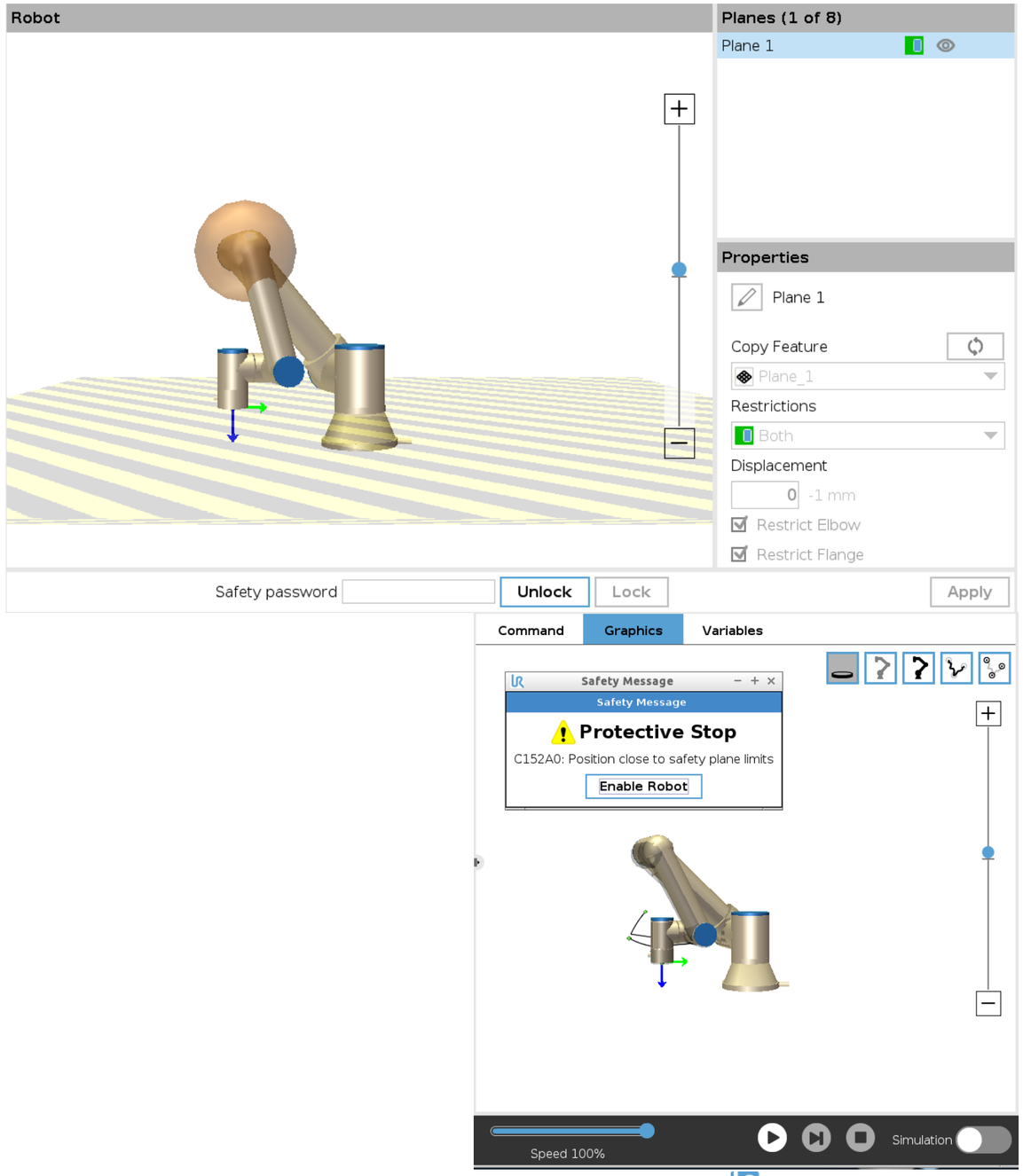
APPLICATION SPECIFIC SINGULARITIES

- Palletizing- when using wizards, it is possible for code to automatically generate waypoints that are out of reach for the robot arm
    - Recommendation- always test out palletizing before implementation

- Force a to reach singularities when in force mode
    - When in force mode- the robot doesn't move to the exact same positions every time, so if you are close to a singularity, your application may become unstable if it hits the singularity
    - Due to fact force mode can lock out different axis, this can lead to unexpected singularities if not properly vetted
    - Recommendation- don’t program waypoints in force mode close to singularities

POSITION LIMITS

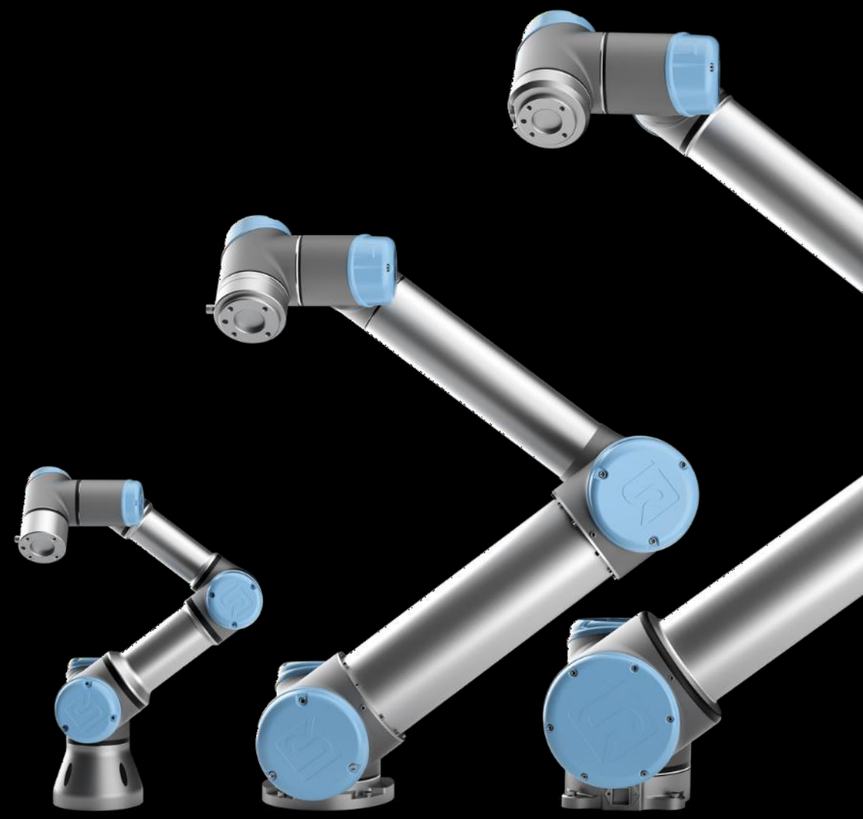
- Robot motion can cause position limit protective stops C150, C151 or C152 during program execution.
- Avoid robot positions close to joint limits or safety limits, like e.g. a safety plane.
- Define the limits according to your risk assessment and ensure a proper margin in the robot program.



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MOTION PROGRAMMING

Protective Stops



UNIVERSAL ROBOTS

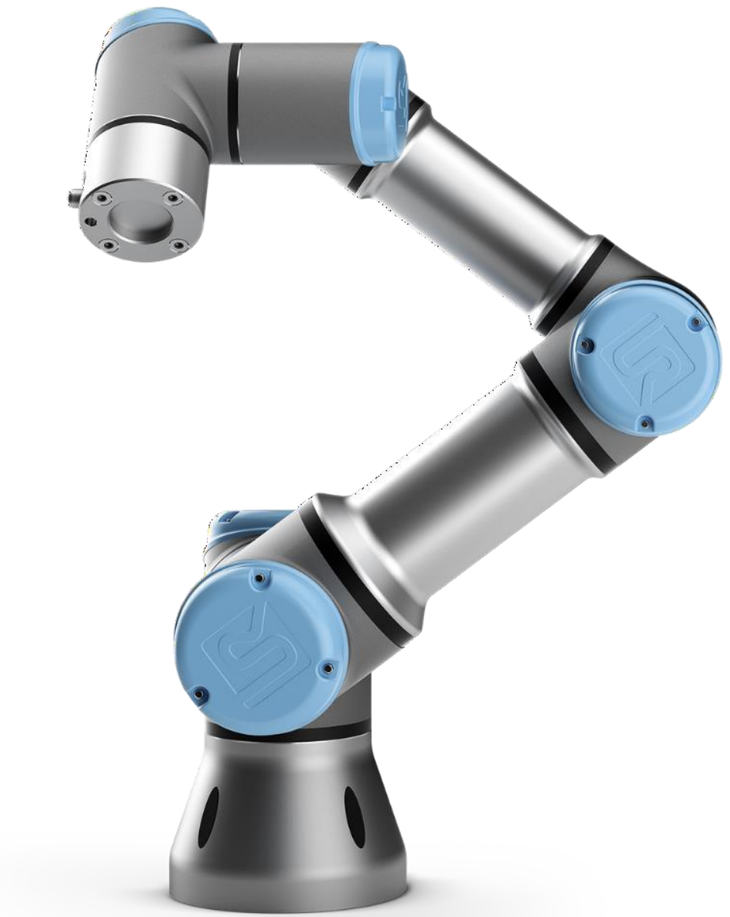
PROTECTIVE STOPS

- The protective stop warning message indicates that the robot program has been stopped because robot cannot follow its intended path

- This functionality is intended to allow the robot to detect external collisions and limit the force it exerts on its environment, protecting any humans sharing its workspace.

- It is not intended to protect the robot from regular collisions with it’s environment.

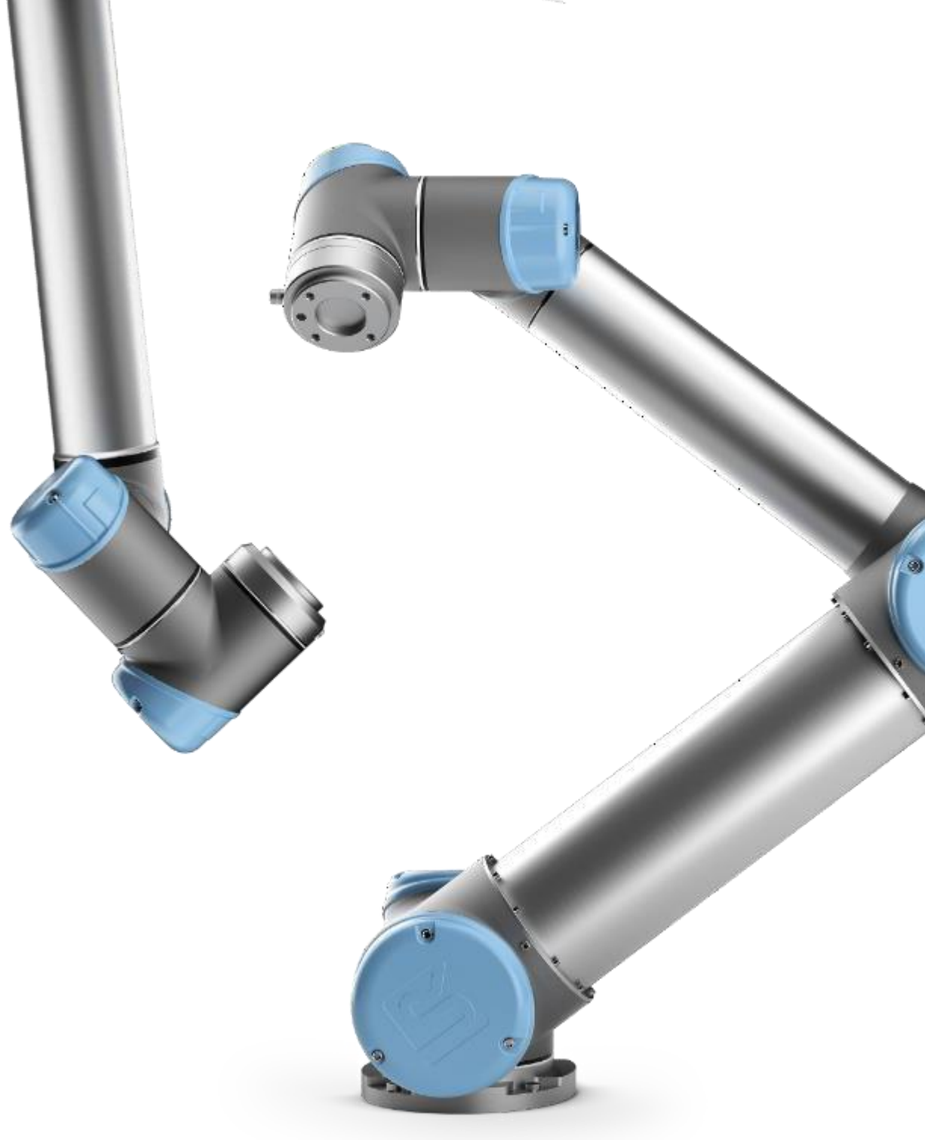
- A well configured application should not experience any protective stops whatsoever in day to day operation.



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PROTECTIVE STOPS

- Protective stops can be caused by any of the following:
  - Incorrect configuration of robot payload mass or centre of gravity
  - Excessive acceleration exceeding the torque generation capabilities of the robot joints
  - Poorly aligned waypoints causing robot tool to collide with workpiece/fixtures.
- Any application experiencing regular protective stops must be adjusted immediately to prevent immature joint failure.



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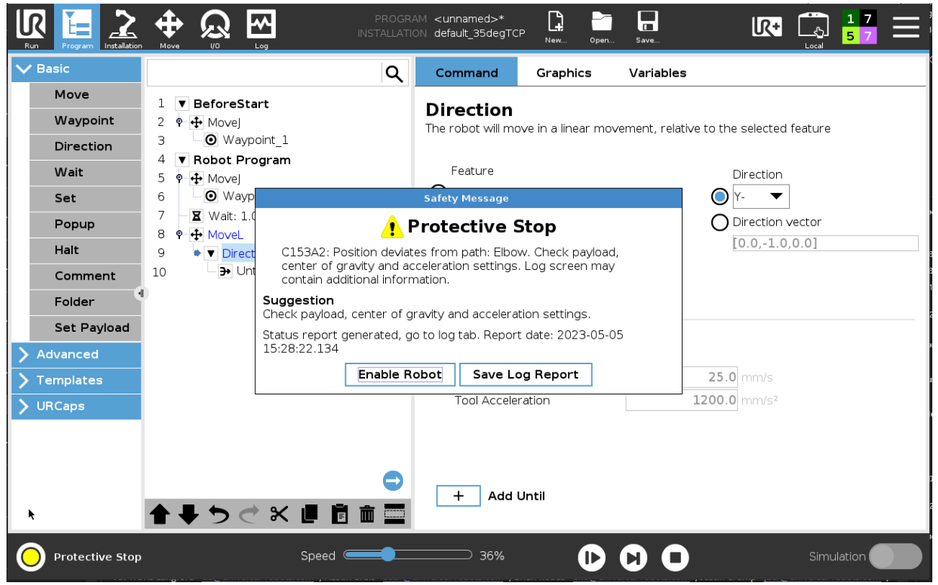
PROTECTIVE STOP CATEGORY

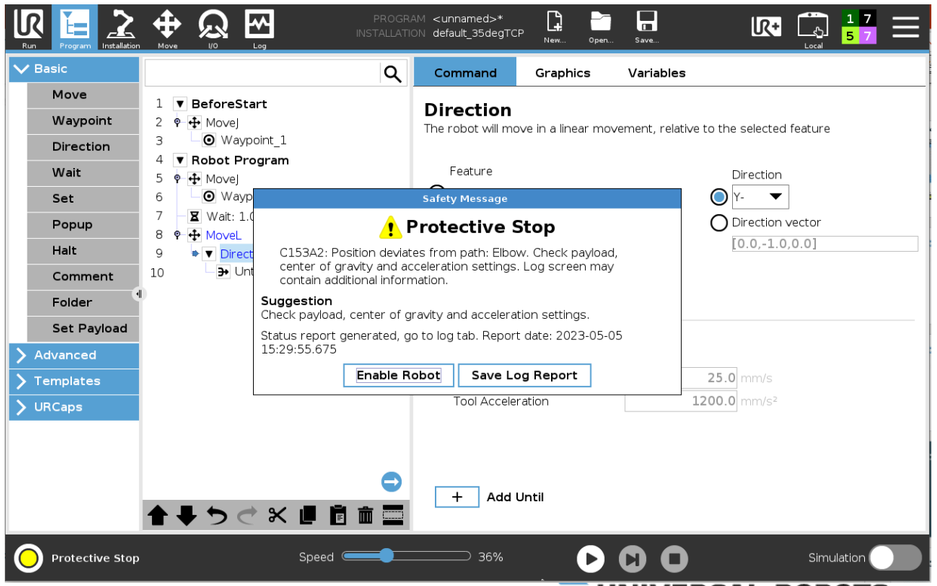
- UR protective stop has a different meaning compared to the same term in safety standards

- Protective stop: describes a way to "protect" against violating a safety limit and "protect" the robot from undue wear and damage.

- There are 2 types of protective stop:
  - One-a resumable (auto reset) from where it was in the program
  - Two-nonresumable
    - If you have too many protective stops within a time window, you cannot resume from that

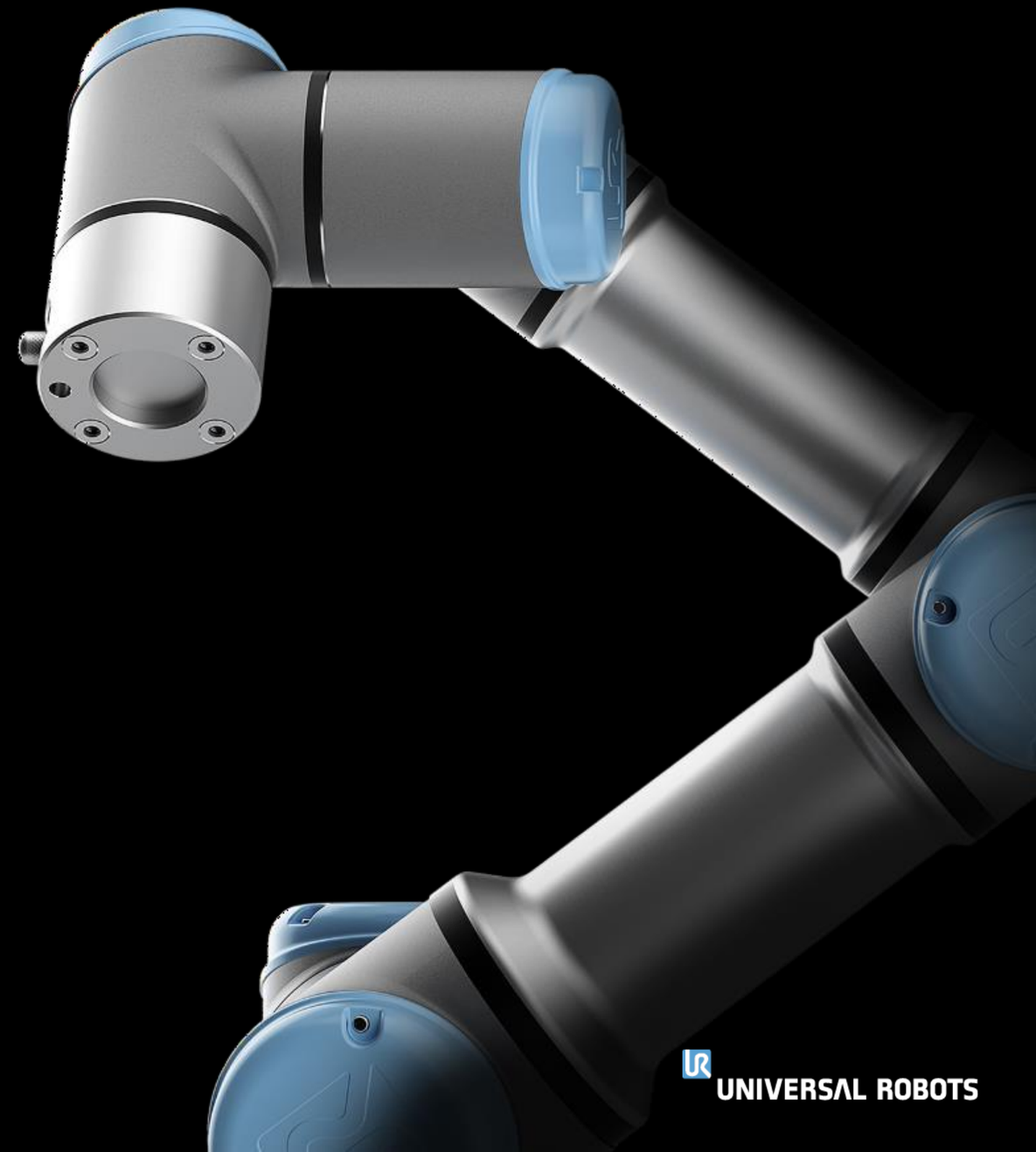
- Resumable p-stop depends on if robot does/does not stop on its original travel path





MAINTENANCE & TROUBLESHOOTING

Recommended Inspection Activities



INSPECTION PLAN (E-SERIES AND NEWER)

- The UR service manual list a number of recommended inspection activities that should be carried out regularly to ensure reliable operation of the system

- On the controller:

Inspection action type			Timeframe		
			Monthly	Biannually	Annually
1	Check Emergency stop on Teach Pendant	F	✗		
2 & 3	On a standard TP: check the freedrive button by pressing it gently. On a 3PE TP: check the 3PE buttons pressing them gently.	F		✗	
	Check Backdrive mode	F	✗		
	Check Freedrive mode	F		✗	
4	Check safety inputs and outputs(if connected)	F	✗		
5	Check Teach Pendant cable and connector	V		✗	
6	Check and clean air filters on Control Box	V	✗		
7	Check terminals in Control Box	F		✗	
8 & 9	Check electrical grounding 1< Ω to Control Box	F			✗
	Check main power to Control Box	F			✗

V = Visual inspection F = Functional inspection



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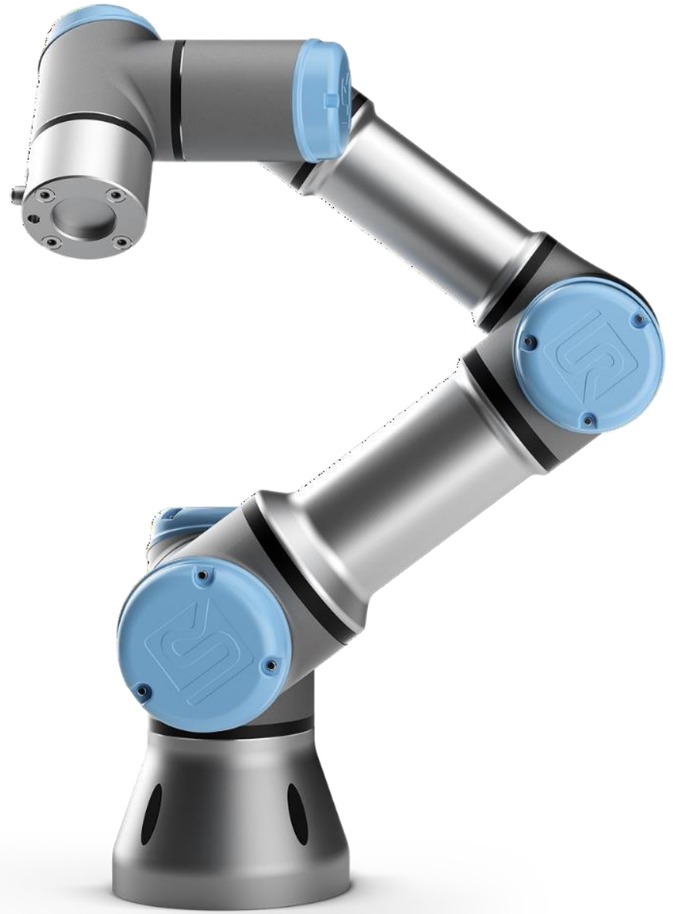
INSPECTION PLAN (E-SERIES)

- On the robot arm:

Inspection action type			Timeframe		
			Monthly	Biannually	Annually
1	Check blue lids *	V		X	
2	Check blue lid screws	F		X	
3	Check flat rings	V		X	
4	Check robot cable	V		X	
5	Check robot cable connection	V		X	
6	Check Robot Arm mounting bolts *	F	X		
7	Check Tool mounting bolts *	F	X		
8	Check screws/bolts on joints *	F		X	

V = Visual inspection F = Functional inspection * = Must also be checked after heavy collision

- If any issues are identified, please rectify immediately or contact your local Universal Robots representative.



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INSPECTION PLAN (UR20/30)

- On the robot arm:

Inspection action type			Timeframe		
			Monthly	Biannually	Annually
1	Check flat rings	V		✗	
2	Check robot cable	V		✗	
3	Check robot cable connection	V		✗	
4	Check Robot Arm mounting bolts *	F	✗		
5	Check Tool mounting bolts *	F	✗		
6	Round Sling	F			✗

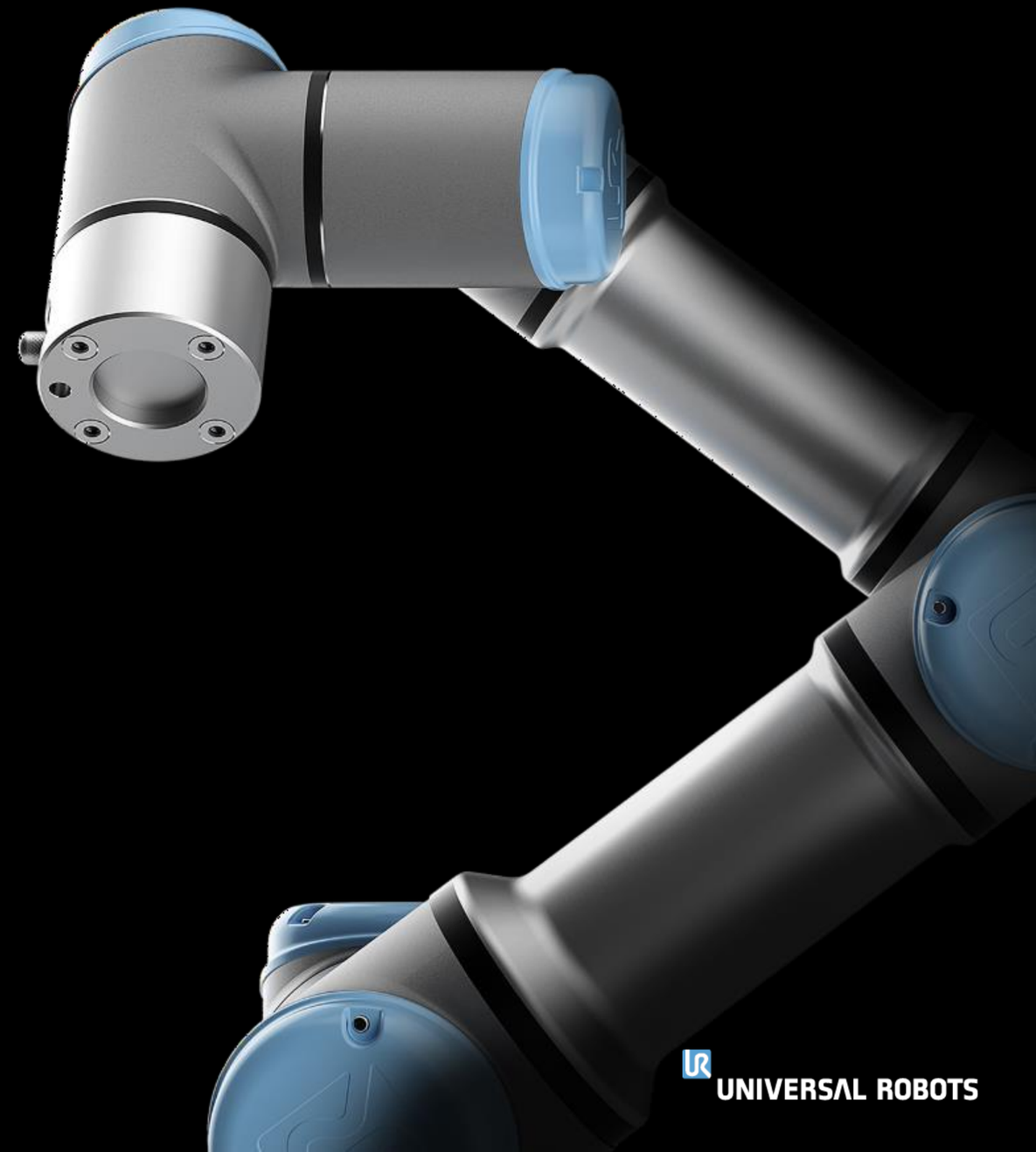
V = Visual inspection F = Functional inspection * = Must also be checked after heavy collision



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MAINTENANCE & TROUBLESHOOTING

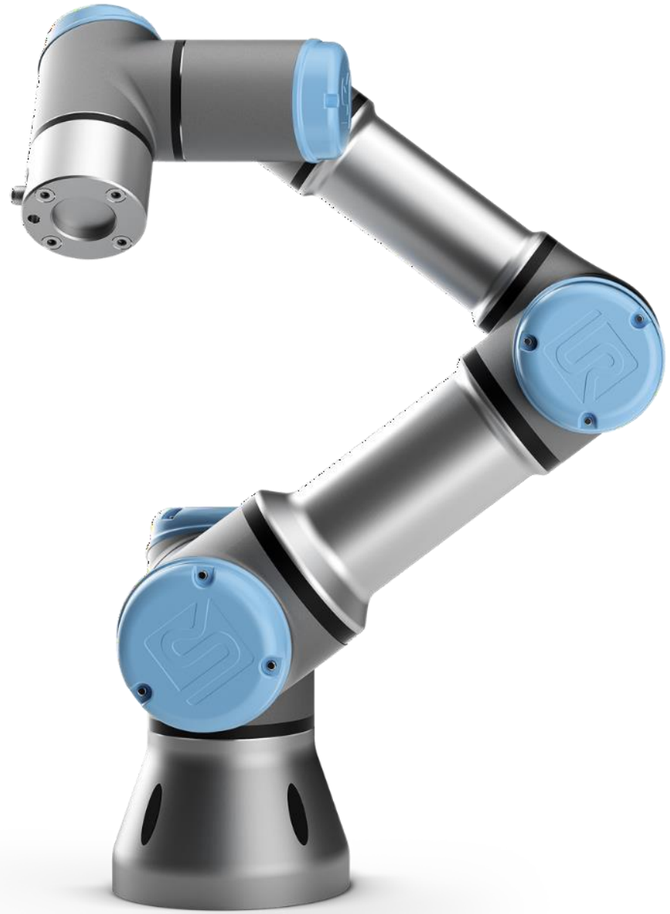
Troubleshooting



POSITION DEVIATION

- The position_deviation_warning() script command allows the user to see when the robot is close to a protective stop.





- Threshold=0 represents no path deviation
- Threshold=1 represents large enough path deviation to trigger a protective stop.

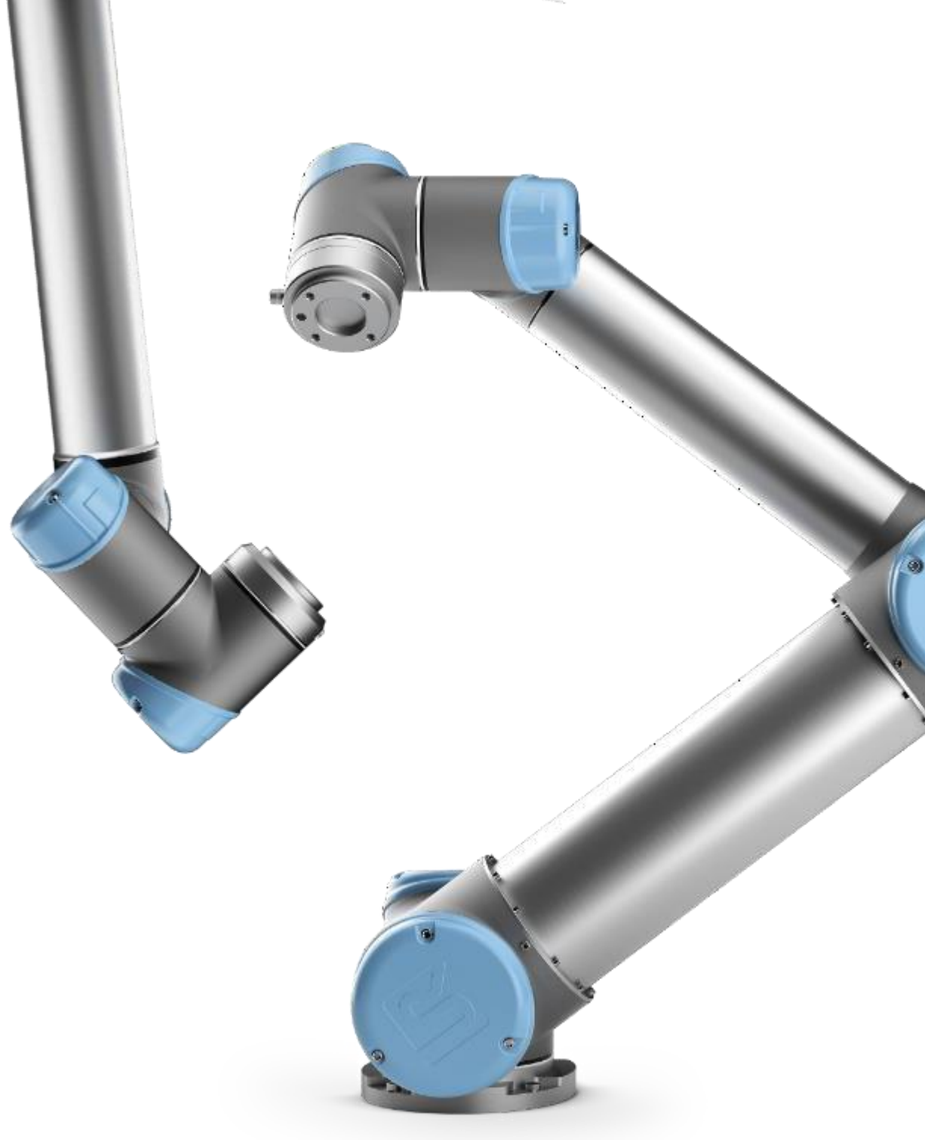
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POSITION DEVIATION

- Add the following script line to the start of your program:
  - position_deviation_warning(True,0.4)

- The first parameter (True) enables position deviation warning messages in the log.

- The second parameter (0.4) defines the threshold that must be exceeded for a warning message to be written to the log.



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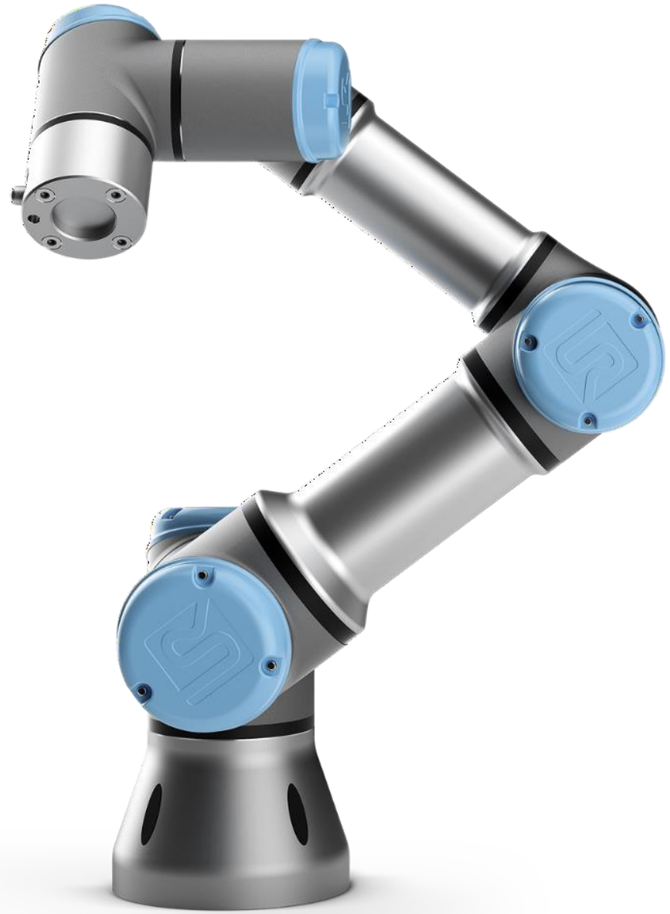
POSITION DEVIATION

- Start with a threshold of 0.4 to check how well your robot is configured to follow it's intended path.

- If the robot deviates by more than 40% of the amount required to trigger a protective stop, a message will be written to the robot log file, specifying which joint is deviating and by how much.

- Identify which movement is causing this deviation, check payload and center of gravity settings and adjust acceleration values for this move until the deviation is sufficiently reduced.

- In a well-tuned application position deviations of more than 20% will not occur.



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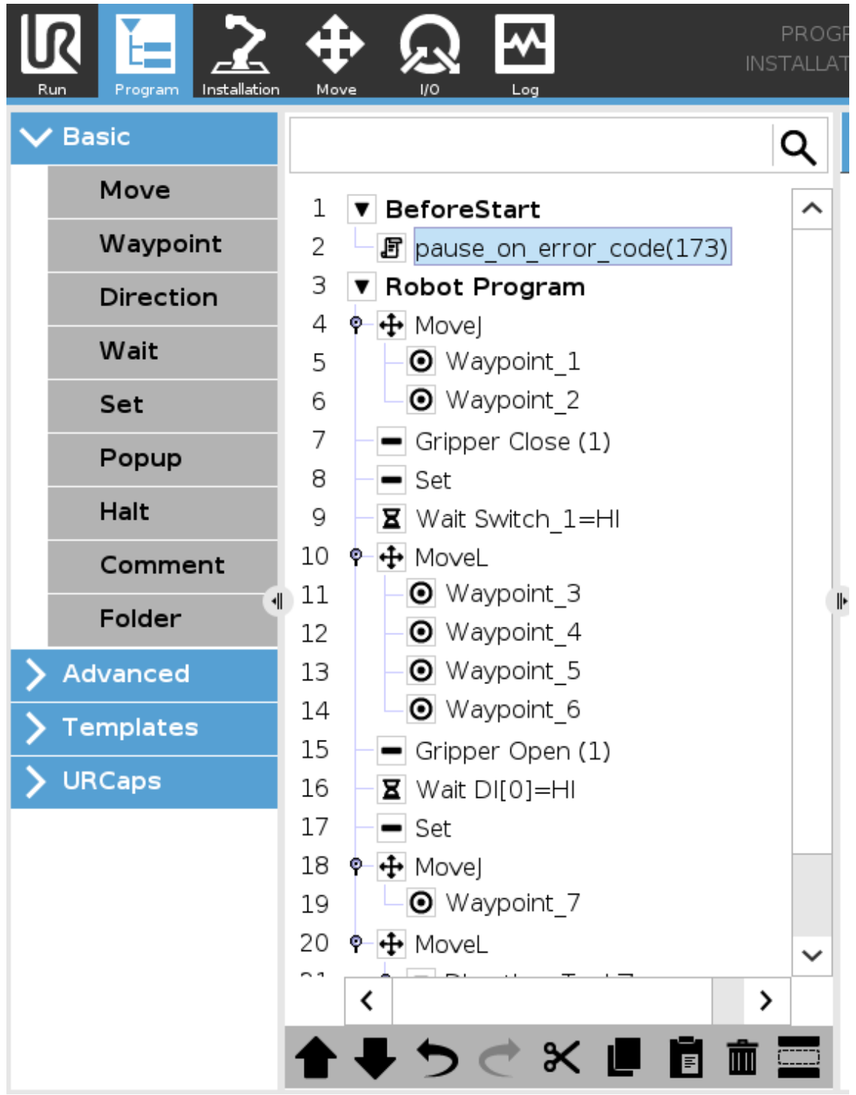
PAUSE ON ERROR CODE

- "pause_on_error_code(x,y)" will pause the program as soon as the specified error occurs. Errors appear on the log as CxAy, where "x" is the code and "y" is the argument. "y" is a completely optional argument

- The function only pauses during program execution and is reset when the program is stopped

- It is recommended to insert this command in the "Before Start" sequence to ensure it is active whenever the "play" button is pressed. Once programs have been debugged, the command can be suppressed or removed

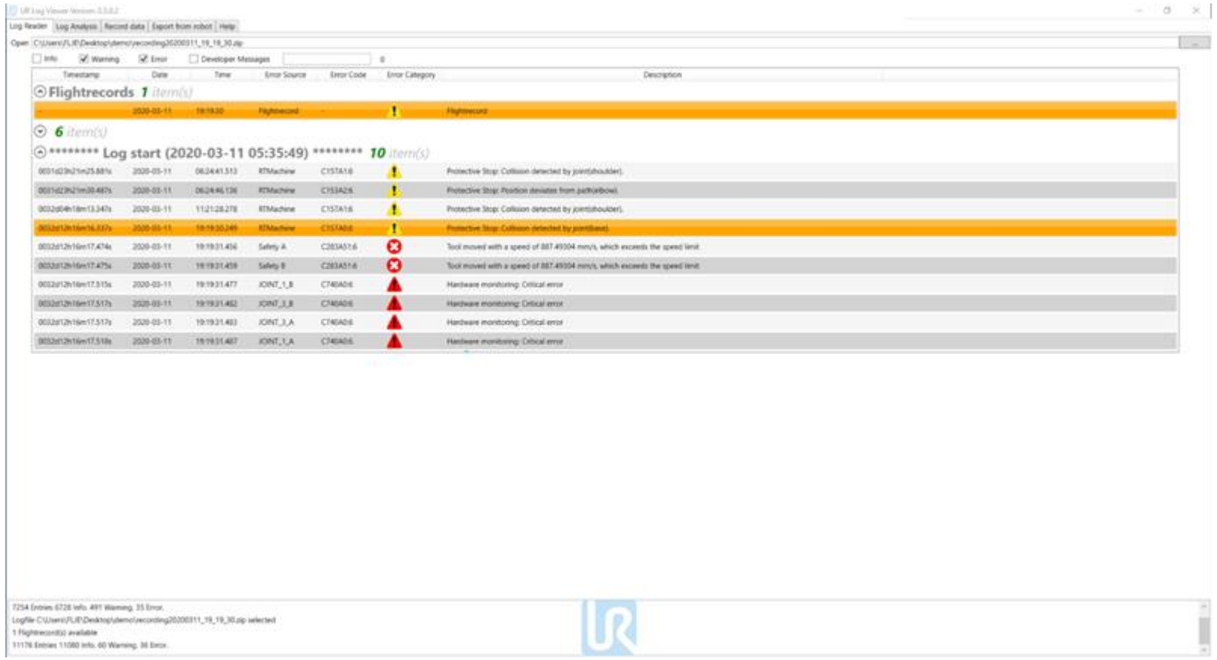
- Typical error codes used are C173/174 relating to "high torque" issues and C171 relating to overlapping blends.



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Log Viewer Tool

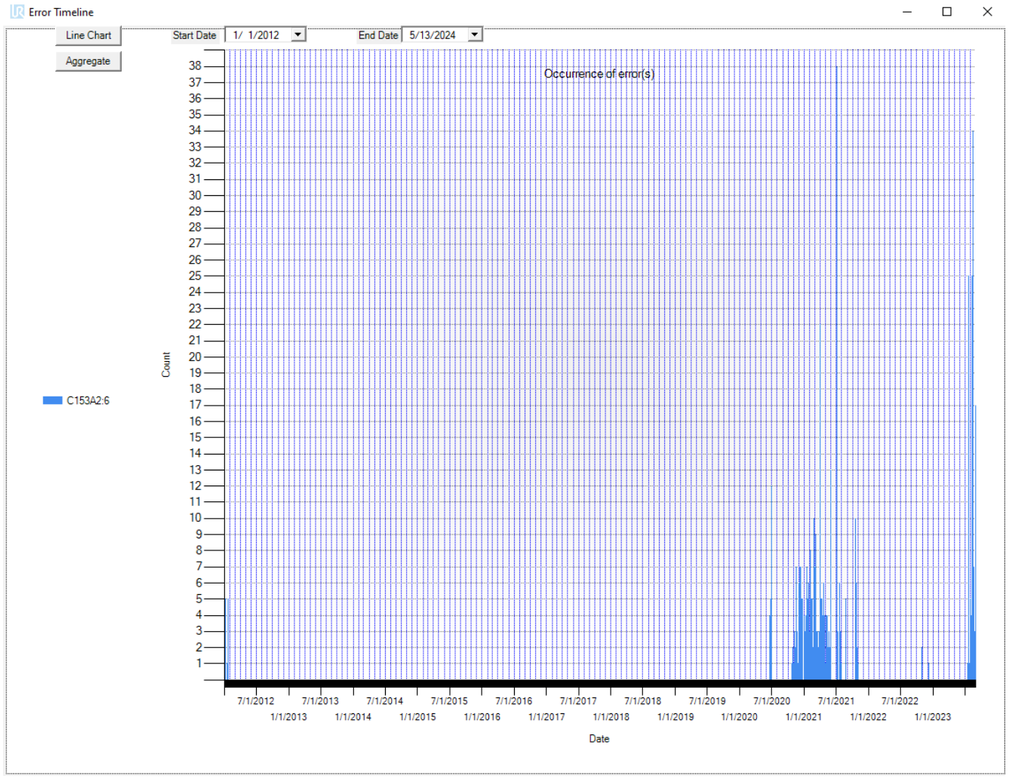
- UR Log Viewer is a Free SW for reading/viewing Support File in order to troubleshoot
- Helps the user understand robot behavior and have data analysis, as well as do improvements to programming/application
- Any findings based on this tool is the sole responsibility and interpretation of the user

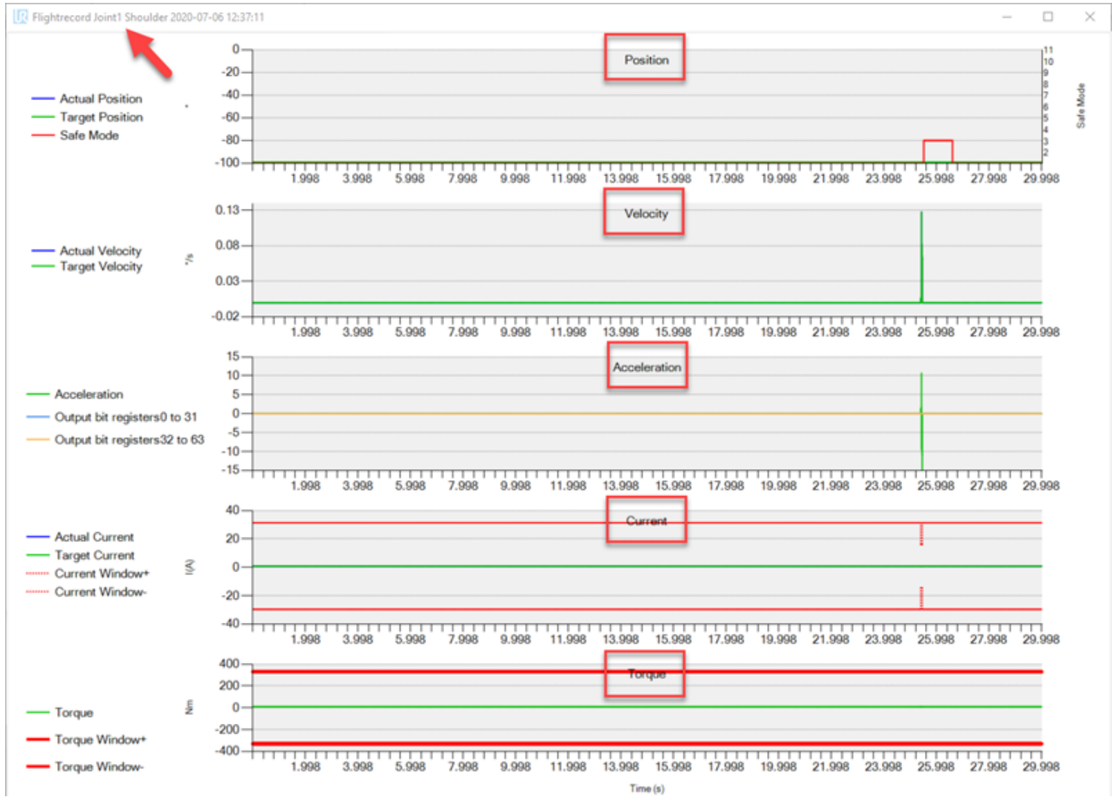


UNIVERSAL ROBOTS

Log Viewer Tool

- User can inspect, visualize, and analyze things such as: log messages and flight records for the entire lifetime of the robot
- Many visualization tools are available
- Can also record data in real-time using Log Viewer Tool





THANK YOU

Please contact your local UR technical support team,
or the distributor from whom you purchased the robot
with any questions.

